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Bachelor Thesis

Modelling Limit Order Books

A data-driven Diffusion Model
in a Markovian Order Book Framework

Franklin Feng Liu

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Supervised by
Prof. Dr. Klaus-Robert Müller (First Examiner)
Prof. Dr. Peter Friz (Second Examiner)
Prof. Dr. Guido Montufár (UCLA)

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F. Liu

Berlin, October 4th 2024 Franklin Liu

Abstract

Limit order books provide real-time digital records of outstanding limit orders for financial securities and provide valuable information about their price formation at a microscopic level. For my bachelor thesis, I consider the active research problem of modelling limit order books. First, I review a recent mathematical framework (Cont et al. 2023 [8]) to model limit order books using a decomposition into a stochastic order flow and a deterministic market clearing operator. Within this framework, I use a data-driven approach with a conditional diffusion model known as TimeGrad (Rasul et al. 2021 [2]) to learn the transition dynamics of order books states over a five second time period using BTC/USDT data. The generated price paths by the diffusion model are able to replicate certain stylized facts like the heavy-tailed distribution of log-returns and the absence of autocorrelation, yet unable to capture the correlation of absolute returns. Additional to the model, a full mathematical derivation of diffusion models in light of the Denoising Diffusion Probabilistic Model (DDPM) approach (Ho et al. 2020 [20]) will be provided with connections drawn to other approaches using stochastic differential equations.

Deutsche Zusammenfassung

Limit-Order Bücher sind digitale Aufzeichnungen von ausstehenden Limit-Orders für Wertpapiere und enthalten Information über deren Preisbildung auf mikroskopischer Ebene. In meiner Bachelorarbeit befasse ich mich mit dem aktuellen Forschungsproblem der Modellierung von Limit-Order Büchern. Zunächst stelle ich einen neuen Ansatz (Cont et al. 2023, [8]) zur mathematischen Modellierung vor, anhand einer Zerlegung in einen stochastischen "order flow" und einen deterministischen Marktoperator. Im Rahmen von diesem Ansatz, stelle ich ein datengetriebenes Diffusion Modell vor, was die Modellarchitektur von TimeGrad (Rasul et al. 2021 [2]) benutzt. Mithilfe des Modells werden die Übergangsdynamiken des Order Buches von BTC/USDT Daten über fünf Sekunden modelliert. Die generierten Preispfade können einige empirische Eigenschaften reproduzieren, wie die Verteilung der log-Renditen und die fehlende Autokorrelation, aber leider nicht die Korrelation der Renditen im Betrag. Zusätzlich zum Modell wird eine vollständige mathematische Herleitung von Diffusionsmodellen mithilfe vom "Denoising Diffusion Probabilistic Model" (DDPM) Ansatz (Ho et al. 2020 [20]) vorgestellt. Darüber hinaus werden Verbindungen zu anderen Ansätzen wie die Nutzung von stochastischen Differentialgleichungen hergestellt.

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1 Introduction

Since the proposal of a central limit order book by the U.S. Securities and Exchange Commission (SEC) in 2000 [41], most major exchanges nowadays like the NASDAQ, New York Stock Exchange and the London Stock exchange employ limit order books [10]. Limit order books (LOBs) are records of outstanding limit orders, which indicate the willingness of market participants to buy or sell a particular financial security at a desired price. These are then used by an order matching engine to execute trades against each other that usually prioritize the price and then the arrival time of orders.

Modelling the dynamics of LOBs is useful for numerous reasons. LOB models can provide an interactive simulation environment for new trading strategies without the need to risk capital [42]. A simulation environment provides an alternative to backtesting on historical data, in which submitted orders do not affect the market in the latter case. This is also a common issue of backtesting, where trading strategies perform well on historical data, but poorly on live data, which is a phenomena that Prata et al. [14] call "simulation-to-reality gap". Additionally, a good understanding of LOBs provides insight into the price formation of securities at a granular level and the market microstructure. For example, Cont et al. [7] provide theoretical and empirical evidence that the so-called "order-flow imbalance", which is a measure of the imbalance in limit buy orders (demand) and limit sell orders (supply) in the order book, has a linear relationship with high-frequency price changes.

However, once a LOB model has been proposed, it raises the the question of what makes a good model. In an attempt to determine a metric for the quality of models, Vyentrenko et al. [42] establish a list of desirable empirical features that LOB models should exhibit. These desirable features are known as "stylized facts" that are statistical properties "common across a wide range of instruments, markets and time periods", which was a concept already highlighted by Cont et al. in 2000 [9]. Many of the stylized facts proposed by Vyentrenko et al. originate from Sherkar and Sen [35], who derived shared empirical features from a total of 378 stocks across 10 different markets and a 9 year period. We will use some of these stylized facts as a guideline to evaluate our model.

This thesis is structured in the following way. Section 2 provides a literature review on mathematical approaches to modelling LOBs, which include agent-based modelling, queuing models and order-driven markets in financial mathematics. Then, we review a general framework proposed by Cont et al. [8], to rigorously define a stochastic order flow and a market clearing operator, which together form an order-book process. A Markov process is used to model the order flow and showcases how models can be constructed in this framework. Section 3 begins with a literature review on data-driven methods of LOB modelling. We propose the use of diffusion models for this task and highlight a motivation of this choice. Before using the diffusion model, a full mathematical derivation of diffusion models from the optimization objective to the training and sampling process is provided. Subsequently, we outline the experimental procedure of training the conditional diffusion model known as TimeGrad [2] on BTC/USDT LOB data to model so-called 'snapshots' of the centered LOB established Section 2. Finally, Section 4 evaluates the diffusion model using stylized facts and Section 5 includes suggestions for future work.

The main contributions of this thesis are:

(1) A broad literature review on various LOB modelling approaches.

(2) The use of diffusion models for LOB modelling, which has not been done in literature to the best of my current knowledge.

(3) The rigorous formulation of the forward process of diffusion models using Markovian transition kernels.

For the sake of completeness, the proofs of known results have been referenced and/or included in the Appendix. Furthermore for better readability, we provide the reference as a footnote of Definitions, Corollaries, Propositions and Theorems found in literature. Otherwise, it can be assumed that these are proposed.

The jupyter notebooks for this thesis can also be accessed here.

Disclaimer: State-of-the-art LOB models are not publicly available, so we rely on the stylized facts as means for evaluating the model.

2 Mathematical Framework for LOBs

Given the complex nature of LOBs, there are many difficulties that arise with modelling, including a trade-off between restrictive assumptions and model complexity. This led to various different mathematical approaches in literature, which we outline in the following.

2.1 Related Work

Typically, a market is composed of different market participants also known as agents, who each have different intentions and strategies. These include market makers, high-frequency traders, institutional traders, trend followers and noise (zero-intelligence) traders [22]. For example, market makers aim to provide liquidity in the markets, whilst maintaining a risk-neutral position. On the other hand, institutional traders may try to minimize their loss incurred by their order impact, when submitting particularly large orders. Therefore, an intuitive approach is to model the strategies of different agents, which is also known as agent-based modelling. The most simple example is the zero-intelligence model, which has been one of the earliest proposed models. As the name suggests, agents do not observe the market and nor have a strategy, meaning they submit orders at random. The viability of zero-intelligence models was studied early on in 2005 by Farmet et al. [15], where they model two types of zero-intelligence agents, who submit limit orders at a frequency determined by two independent Poisson processes with fixed intensity $\mu > 0$ and the price was uniformly sampled in a fixed interval around the mid-price. This model was able to replicate the so-called 'spread' of empirical order book data, which is the difference between the best ask and bid price. However, zero-intelligence is quite a restrictive and controversial assumption. Recently, Cont et al. [11] used a clustering algorithm to segment client order book data, where orders contain the identity of the submitter. Contrary to zero-intelligence, they identified four informed agent clusters, who exhibit similar behavior within the cluster, but differences amongst the clusters in terms of trading frequency, order size and submission times.

The best bid and ask price is also an important feature of the LOB and it is possible to model these quantities in financial mathematics. In financial mathematics, a single-asset financial model is typically assumed to be a filtered probability space $(\Omega, \mathcal{F}_t, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ and a non-negative adapted process $(\bar{S})_{t\in[0,T]} := (R_t, S_t)_{t\in[0,T]}$, where $(R_t)_{t\in[0,T]}$ is a predictable risk-free asset and $(S_t)_{t\in[0,T]}$ is the price process of the considered asset. In this model, there is the assumption that there is no spread, meaning a trader can sell and buy the desired asset at the same price, which is not guaranteed in real markets. Various works in literature such as Jenssen [23], aimed to extend a financial model to an-order driven model, by defining $O_t^{\text{bid}} := (R_t^{\text{bid}}, S_t^{\text{bid}})$ and $O_t^{\text{ask}} := (R_t^{\text{ask}}, S_t^{\text{ask}})$, which form the order-driven financial model $\bar{O}_t := (O_t^{\text{bid}}, O_t^{\text{ask}})$. This order-driven market is also interesting, as Janssen shows that the fundamental theorem of asset pricing (FTAP) can be extended to order-driven markets. The FTAP states that a 'traditional' financial model does not exhibit arbitrage opportunities, if and only if there exists an equivalent martingale measure $\mathbb{P}^*$. Jenssen shows that the order-driven market $\bar{O}_t$ does not allow for arbitrage opportunities, if there exists an equivalent

measure $\mathbb{P}^*$ and adapted process $(S_t^*)_{t\in[0,T]}$ satisfying

$$S_t^{\text{bid}} \leqslant S_t^* \leqslant S_t^{\text{ask}}$$

for all $t \in [0,T]$, such that S^* is a martingale under $\mathbb{P}^*$. It is also possible to extend many other results from financial mathematics to order-driven markets and problems such as portfolio optimization could be regarded in this framework.

Rather than modelling individual agents as in the agent-based approach, one can also model the cumulative order flow as a whole and treat order books as a queuing system. A very popular choice is a Poisson process and numerous models have been proposed in literature with varying intensities [10]. However, since Poisson processes have independent increments, it is difficult to reproduce some stylized facts such as the existence of autocorrelated absolute returns [42]. Therefore, other queuing models such as the Hawkes process [4] have been proposed, which is a so-called self-exciting process, meaning an incoming order influences the intensity of future orders. Given the ambiguity in approaches, Cont et al. [8] developed a very general framework that we will discuss in the following.

2.2 Problem Setting

To get a better idea of a limit order book, Table 1 shows a reconstructed snapshot of the so-called level 3 (L3) limit order book for the Apple stock traded on the NASDAQ that was obtained as a sample from the paid-subscription data-service LOBSTER [21].

Timestamp	Event Type	Order ID	Quantity	Price	Direction
34202.030342281	4	16396051	5	5854600	1
34202.030342281	5	0	77	5854500	1
34202.030451996	-1	16415481	13	5857600	-1
34202.030592674	3	16402646	18	5857900	-1
34202.030597882	3	16402643	18	5857800	-1

Event Type: 1: New limit order, 2: Partial cancellation of order, 3: Full cancellation of order, 4: Execution of visible order, 5: Execution of hidden order
Direction: -1: Sell limit order, 1: Buy limit order

Table 1: Snapshot of Level 3 LOB for AAPL Stock from 2012-06-21 [21]

A limit order is characterized by a timestamp, event type, order ID, quantity, price and direction, which is represented by a row in Table 1. Notice that the second order in Table 1 does not have an order ID. That is because it is a hidden order, which does not appear in the order book, unless it is executed. This order was only visible in Table 1 because LOBSTER reconstructs the order book and infers the hidden orders after execution. Note that not all exchanges allow for hidden orders, but the NASDAQ for example does.

If a buy order (bid) and a sell order (ask) agree at a price, the matching engine executes the trades against each other. Therefore, a market order can be seen as a limit order at the best possible bid/ask. There are also various other forms of limit orders such as "fill and kill" orders, which are orders that either get fully filled immediately or get cancelled.

For the sake of simplicity, we will make some initial assumptions. First, we only consider "good 'til cancelled" orders, which remain in the market until cancelled or fully filled. Furthermore, we only allow limit buy orders, limit sell orders and full cancellation orders. In the following, we take a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ as given. We fix an arbitrary time horizon $T > 0$ and begin with the definition of a limit order.

Definition 2.1 (Limit Order). *A limit order is a vector $x := (x_t, x_i, x_p, x_q)$ with*

(i) arrival time $x_t \in [0, T]$,

(ii) unique order ID $x_i \in \mathbb{R}$,

(iii) order price $x_p \in \delta\mathbb{N} := \{\delta n \colon n \in \mathbb{N}\}$ for some price tick $\delta > 0$,

(iv) order size $x_q \in \mathbb{Z}\backslash\{0\}$, where $x_q > 0$ denotes a sell order and $x_q < 0$ denotes a buy order.

Typically, a trader cannot submit a limit order at any price, but is restricted to some discrete and uniformly spaced price levels. The price increase from one price level to another is known as the tick size $\delta > 0$, which we fix for the rest of the section. Note that the direction of the order is encoded in the order size. Furthermore, a cancellation order is a limit order that uses the same order ID as the original order with the negative order size. This motivates the definition of a L3 order book, where we assume the existence of a market-clearing operator that matches and cancels orders, which we define separately later on.

Definition 2.2 (L3 Limit Order Book[1]). *A L3 limit order book $(L(t))_{t\in[0,T]}$ is a collection of limit orders with $x_t \leqslant t$ for all $x \in L(t)$. Formally, it is a stochastic process taking values in the power set $P := \mathcal{P}([0, T] \times \mathbb{R} \times \delta\mathbb{N} \times \mathbb{Z})$, i.e. $L(t) \colon \Omega \to P$ for all $t \in [0, T]$.*

However, the L3 LOB has quite a complex structure, making it difficult to analyze mathematically. Moreover it is very expensive to obtain L3 LOB data. Nevertheless, some exchanges such as Binance publicly offer a more restrictive LOB that aggregates the quantity of orders at each price level, which is known as the level 2 (L2) limit order book. A snapshot of the L2 order book of Bitcoin/Tether (BTC/USDT), is pictured in Figure 2.

A depth of $n \in \mathbb{N}$ in a L2 LOB refers to the n best (highest) buy orders and (lowest) sell orders. Additionally, the L2 LOB has a simpler structure and we focus on modelling the former and from this point on LOB refers to the L2 LOB. It is possible to derive the L2 LOB from a L3 LOB. For this, we introduce the notion of queue sizes.

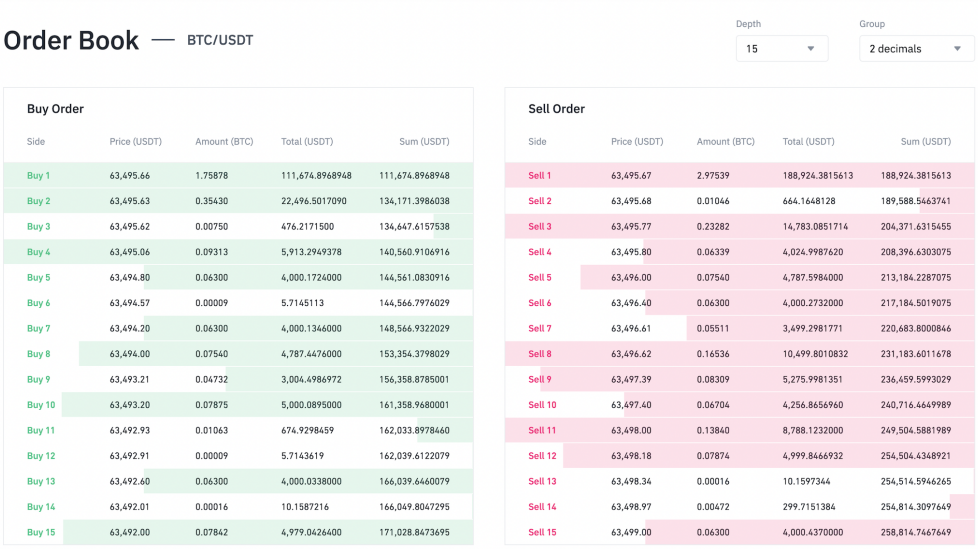
Definition 2.3 (Queue size[2]). *For a L3 LOB $(\mathcal{L}(t))_{t\in[0,T]}$, time $t \in [0, T]$ and price $p \in \delta\mathbb{N}$, let $\mathcal{L}_p(t) := \{x \in \mathcal{L}(t) \colon x_p = p\}$. The quantities*

$$Q_p^-(t) := \sum_{\substack{x\in L_p(t)\\ x_q<0}} -x_q \quad \text{and} \quad Q_p^+(t) := \sum_{\substack{x\in L_p(t)\\ x_q>0}} x_q$$

are called buy queue size and sell queue size, respectively.

[1] Cont et al. *Limit Order Book Simulation with Generative Adversarial Networks*. 2023. pp. 5. https://ssrn.com/abstract=4512356.

[2] See footnote 1



Order Book — BTC/USDT

Depth: 15 | Group: 2 decimals

Buy Order

Side	Price (USDT)	Amount (BTC)	Total (USDT)	Sum (USDT)
Buy 1	63,495.66	1.75878	111,674.8968948	111,674.8968948
Buy 2	63,495.63	0.35430	22,496.5017090	134,171.3986038
Buy 3	63,495.62	0.00750	476.2171500	134,647.6157538
Buy 4	63,495.06	0.09313	5,913.2949378	140,560.9106916
Buy 5	63,494.80	0.06300	4,000.1724000	144,561.0830916
Buy 6	63,494.57	0.00009	5.7145113	144,566.7976029
Buy 7	63,494.20	0.06300	4,000.1346000	148,566.9322029
Buy 8	63,494.00	0.07540	4,787.4476000	153,354.3798029
Buy 9	63,493.21	0.04732	3,004.4986972	156,358.8785001
Buy 10	63,493.20	0.07875	5,000.0895000	161,358.9680001
Buy 11	63,492.93	0.01063	674.9298459	162,033.8978460
Buy 12	63,492.91	0.00009	5.7143619	162,039.6122079
Buy 13	63,492.60	0.06300	4,000.0338000	166,039.6460079
Buy 14	63,492.01	0.00016	10.1587216	166,049.8047295
Buy 15	63,492.00	0.07842	4,979.0426400	171,028.8473695

Sell Order

Side	Price (USDT)	Amount (BTC)	Total (USDT)	Sum (USDT)
Sell 1	63,495.67	2.97539	188,924.3815613	188,924.3815613
Sell 2	63,495.68	0.01046	664.1648128	189,588.5463741
Sell 3	63,495.77	0.23282	14,783.0851714	204,371.6315455
Sell 4	63,495.80	0.06339	4,024.9987620	208,396.6303075
Sell 5	63,496.00	0.07540	4,787.5984000	213,184.2287075
Sell 6	63,496.40	0.06300	4,000.2732000	217,184.5019075
Sell 7	63,496.61	0.05511	3,499.2981771	220,683.8000846
Sell 8	63,496.62	0.16536	10,499.8010832	231,183.6011678
Sell 9	63,497.39	0.08309	5,275.9981351	236,459.5993029
Sell 10	63,497.40	0.06704	4,256.8656960	240,716.4649989
Sell 11	63,498.00	0.13840	8,788.1232000	249,504.5881989
Sell 12	63,498.18	0.07874	4,999.8466932	254,504.4348921
Sell 13	63,498.34	0.00016	10.1597344	254,514.5946265
Sell 14	63,498.97	0.00472	299.7151384	254,814.3097649
Sell 15	63,499.00	0.06300	4,000.4370000	258,814.7467649

Figure 2: L2 LOB Data of BTC/USDT with depth $n = 15$ [5]

Suppose we have a finite amount of price levels $D \in \mathbb{N}$ for an asset, i.e. the asset is bounded by $D\delta$. Then we can define the bid-queue as $Y^-(t) := (Q^-_{1\delta}(t), \dots Q^-_{D\delta}(t))$ and the ask-queue as $Y^+(t) := (Q^+_{1\delta}(t), \dots Q^+_{D\delta}(t))$. We could model a (L2) limit order book as $Y(t) := (Y^-(t), Y^+(t))$ for $t \in [0, T]$ with $Y(t) \in \mathbb{N}^D \times \mathbb{N}^D$, which allows for the clarification of basic terminology. First, define the support operator $\text{supp}(X)$ for $X = (x_1, \dots, x_D) \in \mathbb{R}^D$ as

$$\text{supp}(X) := \{i \in \{1, \dots, D\} \colon x_i > 0\}.$$

Definition 2.4 (Bid, Ask, Spread, Mid-price)**.** *Given some price tick-size $\delta > 0$, time $t \in [0, T]$ and order book queue $Y(t) = (Y^-(t), Y^+(t))$, then*

(i) $b(Y(t)) := \sup\{\text{supp}(Y^-(t))\} \cdot \delta$ *is called the bid price.*

(ii) $a(Y(t)) := \inf\{\text{supp}(Y^+(t))\} \cdot \delta$ *is called the ask price.*

(iii) $m(Y(t)) := \frac{b(Y(t)) + a(Y(t))}{2}$ *is called the mid-price. We also denote this by* $p_{\text{mid}}(t)$*.*

(iv) $s(Y(t)) := a(Y(t)) - b(Y(t))$ *is called the spread.*

We use the convention $\sup(\varnothing) = -\infty$ *and* $\inf(\varnothing) = \infty$*.*

Nevertheless, the mentioned representation $Y := (Y^-, Y^+)$ is computationally inefficient, as $D \gg 0$ and all possible price-level must be included of which many would be 0. Therefore, we use a so-called 'centered' representation[3], where we only consider $d \in \mathbb{N}$ levels around the mid-price. These levels also include more relevant information about the supply and demand and one could make the additional assumption that any orders outside of this $2d$ window

[3]Cont et al. *A Mathematical Framework for Modelling Order Book Dynamics*. 2023, pp. 17. https://arxiv.org/pdf/2302.01169.pdf.

automatically get cancelled. The centered order book takes the form $O(t) := (X(t), p(t))$ and is composed of a price $p(t) := 2p_{\text{mid}}(t) \in \delta\mathbb{N}$ (to ensure that it is a integer) and a queue $X(t) = (X^-(t), X^+(t)) \in \mathbb{N}^{2d+1} \times \mathbb{N}^{2d+1}$ with

$$X^*(t) = \left(Q^*_{\lceil \frac{p_t}{2} \rceil - d\delta}(t), Q^*_{\lceil \frac{p_t}{2} \rceil - (d-1)\delta}(t), \ldots, Q^*_{\lceil \frac{p_t}{2} \rceil}(t), \ldots, Q^*_{\lceil \frac{p_t}{2} \rceil + (d-1)\delta}(t), Q^*_{\lceil \frac{p_t}{2} \rceil + d\delta}(t)\right) \tag{1}$$

for $* \in \{-, +\}$ and $\lceil \frac{p_t}{2} \rceil$ is the mid-price rounded up to the closest tick size. For notational purposes, we re-index the vector from $1, \ldots, 2d+1$ to $-d, \ldots, d$.

As the queue should not be empty as otherwise the mid-price is undefined, the discrete state space of the centered order book O is set as

$$E := \left\{(X, p) = (X^-, X^+, p) \in \mathbb{N}^{2d+1} \times \mathbb{N}^{2d+1} \times \delta\mathbb{N} \colon X^- \neq 0 \text{ and } X^- \neq 0\right\}.$$

Note that the terminology from Definition 2.4 can easily be extended to the the centered order book representation, which we will assume in the following.

A fundamental assumption in mathematical finance is that of no arbitrage. In terms of centered LOBs, there exists arbitrage opportunities however, when $a(X(t)) < b(X(t))$ or equivalently $s(X(t)) < 0$ holds. This can easily be seen, as otherwise a trader could buy an asset at $a(X(t))$ and instantly sell the asset at $b(X(t))$ and collect a profit of $b(X(t)) - a(X(t)) > 0$ per unit. This motivates the definition of an admissible space, which contains all valid configurations of a centered order book.

Definition 2.5 (Admissible space[4])**.** *The set*

$$\mathcal{L} := \{(X, p) \in E \colon a(X) > b(X) \text{ and } a(X) + b(X) = p\}$$

is called the admissible space of LOBs and we say that a centered LOB $O = (X, p)$ is admissible iff $O(t) \in \mathcal{L}$ for all $t \in [0, T]$.

The second condition for the admissible space ensures that p represents the intended quantity of twice the mid-price $2 \cdot p_{\text{mid}}$. We would like to investigate how the centered LOB evolves over time, which is determined by market participants submitting new orders as well as the matching engine executing trades.

2.3 Dynamics of LOB

As the following Corollary shows, incoming orders can be represented by vectors in $\mathbb{R}^{2d+1}$ and order book dynamics can be modelled as a sum of an order and the order book. We denote with $e_i \in \mathbb{R}^{2d+1}$ the unit vector, where the i-th component is 1 and the rest 0.

For notional convenience, we shift the time-index of a process from $X(t)$ to X_t.

[4]Cont et al. *A Mathematical Framework for Modelling Order Book Dynamics*. 2023, pp. 17. https://arxiv.org/pdf/2302.01169.pdf.

Corollary 2.6 (Order Types and Dynamics[5]). *Given a centered order book state* $(X^-, X^+, p) \in E$*, fix an order size* $z > 0$ *and price* $\lceil \frac{p_t}{2} \rceil + i\delta$ *for* $i \in \{-d, \dots, d\}$.

(i) A limit buy order ze_i *causes* $(X^-, X^+) \mapsto (X^- + ze_i, X^+)$.

(ii) A limit sell order ze_i *causes:* $(X^-, X^+) \mapsto (X^-, X^+ + ze_i)$.

iii A cancellation buy order ze_i *causes:* $(X^-, X^+) \mapsto (X^- - ze_i, X^+)$

(iv) A cancellation sell order ze_i *causes:* $i\delta$*:* $(X^-, X^+) \mapsto (X^-, X^+ - ze_i)$

Note that the order book X may not be admissible after new orders enter. This situation could for example occur, if market participants submit limit buy orders at a price higher than or equal to the ask price $a(X)$.

A solution to this problem that Cont et al. [8] proposes, is to decompose the order book into a stochastic order flow and a deterministic clearing function. As an instructive guideline, consider the centered LOB at time $t \in [0, T]$ in some admissible state $O_t \in \mathcal{L}$. As new orders enter the market represented by a process M_t known as the order flow, the order book transitions from (X_t, p_t) to $(X_t + M_t, p_t) \in E$. In a real exchange, the order matching algorithm would then match orders such that the order book returns to an admissible state. We model this using a market clearing operator $\mathcal{C}$ that maps this temporary state back into an admissible state via $\mathcal{C}((X_t + M_t), p_t) \in \mathcal{L}$. This idea is highlighted in the following diagram that can also be found in a similar form in [8].

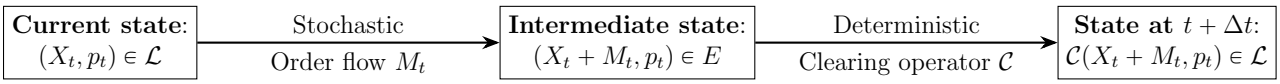


Figure 3: Limit Order Book Decomposition

2.3.1 Markovian Order Flow

We will consider the order flow process $(M_t)_{t\in[0,T]}$ with $M_t \colon \Omega \to E$ in a Markovian setting, as this allows for an elegant description of the centered order book process. This is however not necessary and a more general process may be considered. We begin with the definition of a Markov process on a discrete state space and the transition rate matrix.

Definition 2.7 (Markov Process[6]). *Let* $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ *be a filtered probability space and let* E *be a discrete state space. Then, a* $(\mathcal{F}_t)$ *Markov process is a* $\mathcal{F}_t$*-adapted process* $(M_t)_{t\in[0,T]}$ *with* $M_t \colon \Omega \to E$ *that satisfies*

$$\mathbb{P}[M_t \in B \mid \mathcal{F}_s] = \mathbb{P}[M_t \in B \mid M_s]$$

for all $B \subseteq E$ *and* $s, t \in [0, T]$ *with* $s \leqslant t$.

[5] Cont et al. *A Mathematical Framework for Modelling Order Book Dynamics.* 2023, pp. 4. https://arxiv.org/pdf/2302.01169.pdf.

[6] Andreas Eberle, *Markov Processes* (University of Bonn, 2015), pp. 7. https://wt.iam.uni-bonn.de/fileadmin/WT/Inhalt/people/Andreas_Eberle/MarkovProcesses/MPSkript1415.pdf.

Definition 2.8 (Transition rate matrix[7]). *Let $(M_t)_{t\in[0,T]}$ be a Markov process with discrete state space E, then the transition rate matrix $\rho\colon E\times E\to\mathbb{R}$ is defined for $X,Y\in E$ as*

$$\rho(X,Y)=\lim_{\Delta t\to 0^+}\frac{\mathbb{P}(M_{t+\Delta t}=Y\mid M_t=X)-\delta_{X,Y}}{\Delta t},$$

where $\delta_{X,Y}$ is the Kronecker-delta function and takes the value 1 for $X=Y$ and 0, otherwise.

This is also known as the infinitesimal generator. Given a transition rate matrix ρ with

$$\begin{cases}0 & \leqslant -\rho(X,X)<\infty\\ 0 & \leqslant \rho(X,Y)<\infty \text{ for } X\neq Y\end{cases}$$

and $\sum_{X\neq Y}-\rho(X,Y)=\rho(X,X)$. Schuette and Huisinga (pp. 98, [33]) show that there exists a Markov process $(M_t)_{t\in[0,T]}$ with transition rate matrix ρ, which can be explicitly constructed as follows. Set $T_0=0$ and $M_0=M_{T_0}=X_0$ and recursively define for $k\in\mathbb{N}$

(1) Assume $M_{T_k}=X$.

(2) If $\rho(X,X)=0$ is true, end the construction by setting $M_t=X$ for all $t\geqslant T_k$.

(3) Otherwise choose $\tau_k=\min_{Y\neq X}\{N(X,Y)\}$, where $N(X,Y)$ is exponentially distributed to the parameter $\rho(X,Y)$, if $\rho(X,Y)>0$. Else set $N(X,Y):=\infty$.

(4) Set $T_{k+1}:=T_k+\tau_k$ and

$$\begin{cases}M_t=X & \text{for } t\in(T_k,T_{k+1}),\\ M_t=Y & \text{for } t=T_{k+1},\end{cases}$$

where Y is the state such that $\tau_k=N(X,Y)$.

The above construction also highlights why ρ is called the transition rate matrix, as the elements describe the exponential arrival rate of orders. Cont et al. (pp. 10, [8]) propose to write the transition rate matrix ρ_o of the order flow in terms of arrival rates of buy orders λ_-, sell orders λ_+, cancellation buy orders C_- and cancellation sell order C_+. These are functions of the order price, current state and order size, therefore mapping from $\{-d,\dots,d\}\times E\times\mathbb{N}\to\mathbb{R}$. The transition rate matrix for $X=(X^-,X^+,p)\in E$ is defined as

$$\rho_o(X,Y):=\begin{cases}\lambda_-(i,X,z) & \text{if } Y=(X^-+ze_i,X^+,p),\\ \lambda_+(i,X,z) & \text{if } Y=(X^-,X^++ze_i,p),\\ C_-(i,X,z) & \text{if } Y=(X^--ze_i,X^+,p),\\ C_+(i,X,z) & \text{if } Y=(X^-,X^+-ze_i,p),\\ -\sum_{*\in\{-,+\}}\lambda_*(i,X,z)+C_*(i,X,z) & \text{if } X=Y,\\ 0 & \text{otherwise.}\end{cases}$$

[7] J. Schuette and W. Huisinga, *Markov Chains on Discrete State Space* (Freie Universität Berlin, 2012), pp. 96 https://numerik.mi.fu-berlin.de/wiki/WS_2012/Seminare/Seminar_Stochastik_Dokumente/SchuetteHuisinga_MarkovChainsOnDiscreteStateSpace.pdf.

2.3.2 Market Clearing

The next step is to define a market clearing operator.

Definition 2.9 (Clearing Operator[8])**.** *A clearing operator is a map* $\mathcal{C}\colon E \to \mathcal{L}$*, whose invariant points are* $\mathcal{L}$*, i.e. for* $O \in E$*, we have*

$$O \in \mathcal{L} \quad \iff \quad \mathcal{C}(O) = O.$$

In other words, if the market clearing operator is applied to a non-admissible state $(X, p) \in E \backslash \mathcal{L}$, then we have $(Y, p') := \mathcal{C}(X, p) \in \mathcal{L}$. Recalling the definition of $\mathcal{L}$, this implies the two conditions $a(Y) < b(Y)$ and $a(Y) + b(Y) = p'$ to hold true.

The first condition is independent of p' and we can initially define a market-clearing operator for the queue to satisfy the first condition and then re-center the order book to ensure the second condition. A popular market-clearing method for the queue is so-called order-matching, where limit buy and sell orders are executed against each other according to their price. A precise mathematical definition of this operator and a proof of existence can be found in Proposition 2.3 (pp. 7, [8]). For our purposes, we assume the existence of this order-matching operator $\mathcal{C}'\colon E \to E$ that ensures when applied to any state (X, p), i.e. $\mathcal{C}'(X, p) = (\mathcal{C}'(X), p)$ results in the inequality $b\left(\mathcal{C}'(X)\right) < a\left(\mathcal{C}'(X)\right)$ to hold true. Next, we want to re-center the order book to guarantee the second condition.

Proposition 2.10 (Centering Operator[9])**.** *The centering operator* $J\colon E \to E$ *is defined as* $J(X, p) = (Y, p')$ *with*

$$\begin{cases} p' & = a(X) + b(X), \\ Y & = \sigma_{\lceil \frac{p'}{2} \rceil - \lceil \frac{p}{2} \rceil}(X), \end{cases}$$

where $\sigma_i\colon \mathbb{N}^{2d+1} \times \mathbb{N}^{2d+1}$ *is a shift-operator defined as* $\sigma_i(X) = (\sigma_i^-(X), \sigma_i^+(X))$ *with*

$$\begin{cases} \sigma_i^{\pm}(X)_j & = 0, \quad \text{if } j > \min(d, d-i) \text{ or } j < \max(-d, -d-i), \\ \sigma_i^{\pm}(X)_j & = X^{\pm}_{j+i}. \end{cases}$$

The indices refer to the shifted indices $\{-d, \dots, d\}$ *of the queue size as described in (1).*

It is easily verified that the composition $\mathcal{C} = J \circ \mathcal{C}'$ of the order-matching operator $\mathcal{C}'$ and the centering operator J form a market clearing operator in the sense of Definition 2.9. And we are now ready to define the centered order book.

[8] Cont et al. *A Mathematical Framework for Modelling Order Book Dynamics.* 2023, pp. 5. https://arxiv.org/pdf/2302.01169.pdf.

[9] Cont et al. *A Mathematical Framework for Modelling Order Book Dynamics.* 2023, pp. 19. https://arxiv.org/pdf/2302.01169.pdf.

2.3.3 Centered Order Book

We begin with the definition of a marked point process.

Definition 2.11 (Marked Point Process[10]). *A marked point process on* $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ *is a sequence of random variables* $(t_n, X_n)_{n\in\mathbb{N}}$*, where* $(t_n)_{n\in\mathbb{N}}$ *is an increasing sequence of stopping (or event) times taking values in* $[0,T]$ *with* $t_n \to \infty$ *almost surely as* $n \to \infty$*. Additionally,* $(X_n)_{n\in\mathbb{N}}$ *is a sequence of random variables taking values in some measurable space* E *and* M_n *is* $\mathcal{F}_{t_n}$*-measurable for all* $n \in \mathbb{N}$*. For* $t_n \geqslant T$*, set* $\mathcal{F}_{t_n} = \mathcal{F}$*.*

The condition $t_n \to \infty$ ensures that there are only finite transition times. If the process $(X_n)_{n\in\mathbb{N}}$ is a Markov process, we call this a Markovian marked point process. The centered order book can now be defined as a marked Markovian point process.

Definition 2.12 (Centered Order Book[11]). *Let* $(O_t)_{t\in[0,T]}$ *be a Markovian order flow process with transition rate matrix* $\rho_o \colon E \times E \to \mathbb{R}$ *and clearing operator* $\mathcal{C} \colon E \to \mathcal{L}$*.*

A centered order book is a Markovian marked point process $(X_n, p_n)_{n\in\mathbb{N}}$ *with* $X_n \in \mathcal{L}$ *and defined as follows. Suppose* $t_1, \ldots, t_N \in [0,T]$ *is a sequence of event times with corresponding states* $(X_1, p_1), \ldots, (X_N, p_N)$*, where* (X_k, p_k) *is the state of the centered order book during* $[t_{k-1}, t_k)$ *and* $t_0 = 0$*. Then, for all* $k \in \{1, \ldots, N\}$

(i) $t_k - t_{k-1}$ *is defined to be exponentially distributed according to the rate* $\sum_{Y\in E} \rho_0\left((X_{k-1}, p_{k-1}), Y\right)$*.*

(ii) $(X_k, p_k) \in \mathcal{L}$ *is distributed according to the probability*

$$\frac{\sum_{Y\in\mathcal{C}^{-1}(\{(X_k,p_k)\})} p_o\left((X_{k-1}, p_{k-1}), Y\right)}{\sum_{Y\in E} \rho_o\left((X_{k-1}, p_{k-1}), Y\right)},$$

where $\mathcal{C}^{-1}(\{(X_k, p_k)\})$ *denotes the pre-image of the clearing operator.*

Given poperty (ii), the process $(X_n, p_n)_{n\in\mathbb{N}}$ is indeed a marked Markovian point process.

2.4 Example

We will show an example using the defined framework. Cont et al. [8] suggests the transition rates to be functions of the distance to the current bid and ask price. For simplicity, only orders of size $z = 1$ are considered. The transition rates for limit buy λ_- and sell λ_+ orders, parameters $\mu, \beta > 0$ and tick-siez $\delta > 0$ are defined in the following way

$$\lambda_-(k, X, z) := \begin{cases} \frac{\beta}{(\frac{a(X)}{\delta} - k)^\alpha} & \text{if } k < \frac{a(X)}{\delta}, z = 1, \\ \mu & \text{if } k = \frac{a(X)}{\delta}, z = 1, \\ 0 & \text{otherwise,} \end{cases} \quad \lambda_+(k, X, z) = \begin{cases} \frac{\beta}{(k - \frac{b(X)}{\delta})^\alpha} & \text{if } k > \frac{b(X)}{\delta}, z = 1, \\ \mu & \text{if } k = \frac{b(X)}{\delta}, z = 1, \\ 0 & \text{otherwise.} \end{cases}$$

[10] Peter Tankov and Rama Cont, *Financial Modelling with Jump Processes*, Chapman & Hall/CRC Financial Mathematics Series. 2003, pp. 74.

[11] Cont et al. *A Mathematical Framework for Modelling Order Book Dynamics*. 2023, pp. 11. https://arxiv.org/pdf/2302.01169.pdf.

Thus limit orders further away from the best-bid and best-ask have a lower arrival rate. Additionally, the cancellation bid C_- and ask C_+ transition rates are given by

$$C_-(k, X, z) = \begin{cases} \theta(\frac{b(X)}{\delta} - k)X_k^- & \text{if } k < \frac{b(X)}{\delta}, z = 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$C_+(k, X, z) = \begin{cases} \theta(k - \frac{a(X)}{\delta})X_k^+ & \text{if } k > \frac{a(X)}{\delta}, z = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where $\theta\colon \mathbb{N} \to \mathbb{R}$, $i \mapsto \theta(i)$ is a function of the price $i \in \{1, \dots, d\}$ and X_k^-, X_k^+ refer to the bid and ask queue size at price k respectively.

Using Definition 2.12 of the centered order book, the order book dynamics could be simulated by computing the transition rates $\rho_o(X, Y)$ based on the described rates. Following this, the next event time can be sampled from the prescribed exponential distribution along with the corresponding order book state.

3 Data-driven Modelling of LOBs

Using the centered order book framework that was introduced in Section 2 as a guideline, we will develop a data-driven model using a conditional diffusion model in this section. We begin with a literature review of existing work, followed by an introduction to diffusion models and finally use an auto-regressive conditional diffusion model known as TimeGrad [2] to model the dynamics of so-called 'snapshots' of a centered order book, which represent consecutive order book states sampled at uniformly-spaced time periods.

3.1 Related Work

Given the abundance of financial data, machine learning has naturally become a widespread industry and academic research tool. Machine learning in regards to modelling LOBs have been explored through two main approaches in literature, which was discussed in the recent review of LOB simulations by Jain et al. [22]. On one hand, there are implicit models of LOB dynamics using downstream tasks such as price trend classification. And on the other hand, there are explicit models that attempt to model the LOB itself.

To address the first approach of implicit models through downstream tasks, there has been extensive research in the supervised classification task of predicting the label of a price trend using the LOB as data, which is known as the stock price trend prediction (SPTP) problem. Fundamentally, this task assigns the label 'upward', if the mid-price at a fixed time horizon in the future increases above some threshold, 'downward' for a decrease below a threshold and 'stable', if it remains between the threshold. We refer to [14] for an exact problem formulation. Various models such as multi-layer perceptions and support vector machines [30] have been proposed in literature. Notably more advanced architectures such as DeepLOB [43], which include convolutional layers and long short-term memory modules (LSTMs), are able to achieve remarkably high degrees of classification accuracy. In effort to provide comparative grounds for such models, Ntkaris et al. released a benchmark LOB dataset [1] named FI-2010 using five stocks of the NASDAQ Nordic exchange. Consequently, this classification task is being actively researched in literature and interest was recently enforced by a benchmark study from JP Morgan researchers [14], who compared various of these models on different datasets and released an open-source python framework known as LOBCAST, which allows for easier production of SPTP models and benchmark comparisons.

This classification problem has also been explored from another perspective, as Sirignano and Cont [36] investigated the existence of a so-called "universal price mechanism", which provides insight into the price formation that is supposedly independent of specific assets. At a more abstract level, this can be formulated as the following problem: Given a universe of assets $(S^i)_{i\in\mathcal{I}}$ for some index set $\mathcal{I}$ and price processes $(S^i_t)_{t\in[0,T]}$, does there exist a function F such that small price changes in the future denoted by Δt can be expressed as a function of the current state, i.e.

$$S^i_{t+\Delta t} = F(X^i_t, \varepsilon_t),$$

where X^i_t represents a state variable that may include any information such as price history that is known up to time t and ε_t is a noise term. In order to test this hypothesis, Sirignano

and Cont considered an almost identical classification task and trained a recurrent neural network on LOB data acting as state variables X_t. They found that the model trained on a universe of assets was able to outperform the same model trained only using a single-asset, which suggests possible similarities in the price formation amongst a wide range of assets.

In terms of explicit models of order books also known as synthetic generation of LOBs, there have also been different approaches in literature, which mainly make use of generative adversarial networks (GANs) as a training paradigm. GANs were introduced 2014 by Goodfellow et al. [18] and are based on a 2-player minimax game, where one agent, known as the generator, attempts to generate new samples of the underlying data distribution, in order to 'fool' the other agent, known as the discriminator, which attempts to distinguish the generated samples from actual samples. For example, Prenzel et al. [31] use a conditional GAN to dynamically generate order intensities of the order flow as described in Section 2.3.1 and were able to replicate the unconditional distribution of the intensities in its generated order book process. Another approach is to model the order book messages or the L3 LOB (see Definition 2.2), whereby Nagy et al. [28] propose a large language model that tokenize a sequence of order book messages and learn the conditional probabilities of order book messages. However, this granular approach of modelling the low-level messages of an order book has only been introduced recently and there is not much literature in this area yet.

Lastly and consistent with the centered order book framework introduced in Section 2.3.3, Cont et al. [12] propose a Wasserstein GAN to model snapshots of the centered order book. Specifically, they learn the conditional distribution of consecutive order book states separated by 10-second intervals. Their model is trained on 20 days worth of order book data of the large-tick stock Micron Technology (MU) and is able to reproduce the marginal distributions and correlations between the order book levels. Furthermore, the model can generate realistic price-paths that satisfy empirical properties like market order impact, which indicate how submitted orders affect the price and was studied extensively in [7]. This suggests viability of this model to be used as a simulator. However, in the future research section, Cont et al. [12] point out the problem with GANs. GANs are known to suffer from convergence issues due to its adversarial training as well as so-called mode collapse, where the model is only able to generate a limited variety of output that may be very similar, which may not be suitable for generating price-paths, due to its high volatility. Therefore, we propose the use of diffusion models, which have gained a lot of popularity in recent years and are generally stable during training and do not suffer from mode collapse.

3.2 Diffusion Models

3.2.1 Introduction to Diffusion Models

One of the most influential papers in machine learning was the introduction of diffusion models by Sohl-Dickstein et al. in 2015 [37], which was originally motivated by thermodynamics. Yet unbeknownst of the power and flexibility of diffusion models, it was only in 2020, when Ho et al. [20] showcased the ability of diffusion models for image generation on the benchmark image dataset CIFAR10. Since then, diffusion models like stable diffusion [32] have become an industry standard, producing state-of-the-art results, even surpassing GANs [13] on the Fréchet inception distance score, which is a metric to evaluate the quality of generated samples. In current literature, diffusion models have been formulated in three different ways, namely denoising diffusion probabilistic methods (DDPM) [20], score-based generative modelling [38] and using stochastic differential equations [39]. Since the model we use is based on the DDPM approach, we consider diffusion models through the scope of DDPM and draw connections to the other approaches in Section 3.2.3.

3.2.2 Denoising Diffusion Probabilistic Methods (DDPM)

The aim of diffusion models is to generate new samples from some underlying data distribution Q on $\mathbb{R}^d$. In the initial formulation of DDPMs [20], this is achieved by transforming the data distribution Q into a standard Gaussian distribution $\mathcal{N}(0, \mathrm{I}_d)$ through a forward/noising process. More specifically, given some datapoint $X_0 \sim Q$, we iteratively perturb X_0 with a series of i.i.d. Gaussian noise. Following this, the model learns a reverse process, in order to reconstruct the original data from the perturbed data and therefore learns the reverse transformation from the standard Gaussian distribution back to Q. The model is then able to generate new samples using this reverse process by simply taking Gaussian noise as input and denoising it. This idea is illustrated in the following figure.

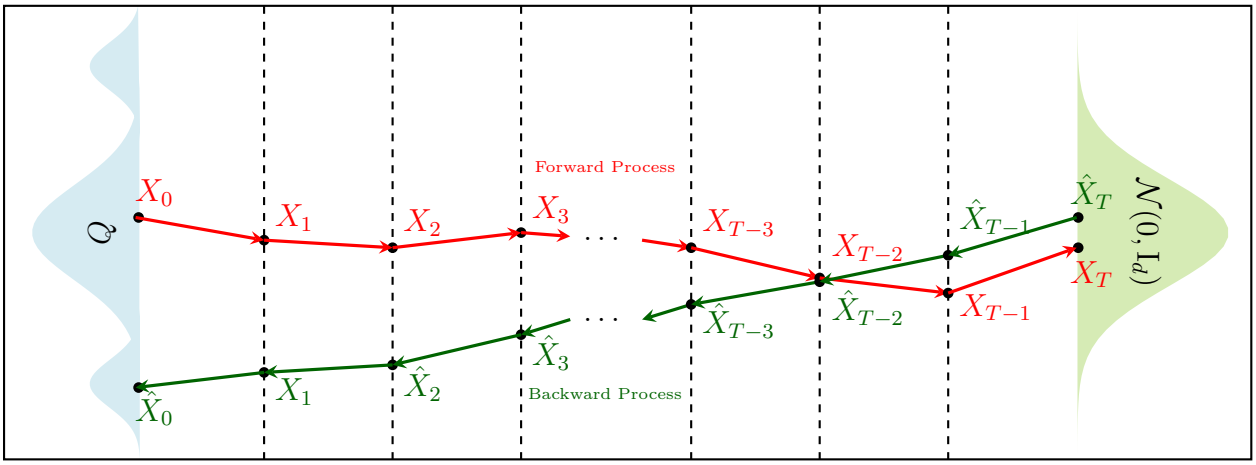


Figure 4: Forward and Backward process of Diffusion model

Before delving into a detailed mathematical derivation of diffusion models, we first provide an instructive guideline of the DDPM approach of diffusion models at a high-level:

(1) **Defining Objective**: The goal is to minimize some metric between the true data distribution Q and model distribution P_θ, which are assumed to have densities q and p_θ. This turns out to be equivalent to minimizing the negative log-likelihood $\mathbb{E}_q[-\log p_\theta]$.

(2) **Forward Process**: For a given datapoint $X_0 \sim Q$ with $X_0 \in \mathbb{R}^d$, define a forward/noising process $X_0, \dots, X_T$ using Gaussian transition kernels. Show that for any initial data distribution Q, the forward process converges in distribution against the standard Gaussian distribution, i.e. $X_T \xrightarrow{\mathrm{d}} \mathcal{N}(0, \mathrm{I}_d)$ as $T \to \infty$. Therefore, we make the assumption that the model distribution of the final step is standard Gaussian, i.e. $p_\theta(x_T) = \mathcal{N}(0, \mathrm{I}_d)$.

(3) **Backward Process**: Using the forward process, a tractable upper bound on the negative log-likelihood is derived and minimized. The solution of this optimization task yields a model backward process $p_\theta(x_{t-1} \mid x_t)$, which can then be used to sample from $p_\theta(x_0)$, using the joint distribution

$$p_\theta(x_0, \dots, x_T) = p_\theta(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t),$$

which is true as the denoising process $X_T, \dots, X_0$ is also a Markov process. Under the assumption that $p_\theta(x_T) = \mathcal{N}(0, \mathrm{I}_d)$ holds true, we first sample $x_T \sim \mathcal{N}(0, \mathrm{I}_d)$ and iteratively sample from the model conditionals $p_\theta(x_{t-1}|x_t)$ for $t = T, \dots, 1$.

Defining Objective

First, we illustrate the mathematical problem that diffusion models attempt to solve. In the following, we assume a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and let $\mathcal{M}_1^+(\mathbb{R}^d)$ denote the set of probability measures on $\mathbb{R}^d$ with a Lebesgue density. Let $Q \in \mathcal{M}_1^+(\mathbb{R}^d)$ be the ground-truth data distribution with density $q(x)$ and let $\{P_\theta \in \mathcal{M}_1^+(\mathbb{R}^d) \colon \theta \in \Theta, \mathrm{d}P_\theta = p_\theta(x)\mathrm{d}x\}$ be the model space for some parameter space $\Theta \in \mathbb{R}^k$, $k \in \mathbb{N}$. In the context of machine learning, the parameter space typically refers to the weights of a neural network.

The aim is to find the parameter that minimizes the distance between the true distribution and model distribution with respect to some distance measure $\mathbf{d}$ on probability measures, i.e. $\operatorname{argmin}_{\theta \in \Theta} \mathbf{d}(Q, P_\theta)$. A commonly used distance measure on probability measures is known as the Kullback-Leibner (KL) divergence (pp. 35, [19]).

Definition 3.1 (Kullback-Leibner Divergence)**.** *Let μ, ν be probability measures on $\mathbb{R}^d$. If ν is absolutely continuous with respect to μ, the Theorem of Radon-Nikodym*[12] *provides the existence of a density $\frac{\mathrm{d}\nu}{\mathrm{d}\mu}$ and the quantity*

$$\mathbf{d}_{\mathrm{KL}}(\mu \| \nu) = -\int_{\mathbb{R}^d} \log\left(\frac{\mathrm{d}\nu}{\mathrm{d}\mu}\right) \mathrm{d}\mu.$$

is known as the KL-divergence between μ and ν. Furthermore, if μ is not absolutely continuous with respect to ν, we set $\mathrm{d}_{\mathrm{KL}}(\mu \| \nu) = \infty$.

[12] William Feller, *An Introduction to Probability Theory and Its Applications*, vol. 2 (New York: John Wiley & Sons, 1966), pp. 140.

Note that the KL-divergence is not a metric, as one can construct conter-examples to show that it is not symmetric. However, it can be shown that is positive definite, which is sufficient for a notion of distance. In our case, where Q and P_θ are assumed to have Lebesgue densities $p_\theta(x)$ and $q(x)$ respectively, it can be easily verified that if Q is absolutely continuous w.r.t. P_θ, then the function $\frac{p_\theta(x)}{q(x)}$ defined Q-almost everywhere corresponds to the the Radon-Nikodym density $\frac{\mathrm{d}P_\theta}{\mathrm{d}Q}$. In particular, the KL-divergence is given by

$$\mathrm{d_{KL}}(Q\|P_\theta) = -\int_{\mathbb{R}^d} \log\left(\frac{p_\theta(x)}{q(x)}\right) q(x)\mathrm{d}x. \tag{2}$$

Instead of minimizing the KL-divergence between Q and P_θ, the following Corollary shows that this is equivalent to minimizing the so-called negative log-likelihood.

Corollary 3.2. *Let $Q \in \mathcal{M}_1^+(\mathbb{R}^d)$ be a probability measure on $\mathbb{R}^d$ with density $q(x) \in \mathcal{L}^1(\mathbb{R}^d)$. Furthermore, let $(P_\theta)_{\theta\in\Theta} \subseteq \mathcal{M}_1^+(\mathbb{R}^d)$ be a family of probability measures on $\mathbb{R}^d$ parameterized by a parameter space $\Theta \subseteq \mathbb{R}^k$ with densities $p_\theta(x)$ for all $\theta \in \Theta$. Then the following relation holds true*

$$\mathrm{argmin}_{\theta\in\Theta}\, \mathrm{d_{KL}}(Q\|P_\theta) = \mathrm{argmin}_{\theta\in\Theta} -\int_{\mathbb{R}^d} \log p_\theta(x) q(x)\mathrm{d}x$$

as long as both sides are well-defined. The quantity $-\int_{\mathbb{R}^d} \log p_\theta(x)q(x)\mathrm{d}x$ is known as the negative log-likelihood, which we denote as $\mathbb{E}_q[-\log p_\theta]$.

Proof. The proof follows immediately from equation (2), since we have

$$\mathrm{d_{KL}}(Q\|P_\theta) = -\int_{\mathbb{R}^d} \log\left(p_\theta(x)\right) q(x)\mathrm{d}x + \int_{\mathbb{R}^d} \log\left(q(x)\right) q(x)\mathrm{d}x$$

and the second term on the right hand side does not depend on the parameter θ. □

For a family of known distribution $(p_\theta(x_0))_{\theta\in\Theta}$, the negative log-likelihood can be computed empirically using the data. However, if we use neural networks, often times p_θ and therefore the negative log-likelihood is not tractable. In this case, it would be advantageous to have a tractable tight bound on the negative log-likelihood. In order to derive a bound, we will first define the forward process of the diffusion model.

Forward Process

In most literature on diffusion models, authors define the forward process by specifying the conditional probabilities. However, when dealing with conditional probabilities of continuous random variables X and Y, one has to be more cautious, as the definition $P(X \in B|Y = k) = \frac{P(X\in B, Y=k)}{P(Y=k)}$ does not make sense, when $P(Y = k) = 0$. In particular, when Y is continuous and for example normally distributed, this is always the case. This problem can be overcome by defining conditional probability densities[13] $p(x|y) := \frac{p(x,y)}{p(y)}$, but requires the

[13] William Feller, *An Introduction to Probability Theory and Its Applications*, vol. 2 (New York: John Wiley & Sons, 1966), pp. 72.

existence of a joint distribution $p(x, y)$. In general, if X and Y both have densities, this does not guarantee the existence of a joint density $p(x, y)$. We loosen this assumption, by using Markov transition kernels that impose measurabilty properties, in order to show the existence of a joint distribution for the forward process and provide a rigorous mathematical framework for diffusion models. This framework has a number of advantages, as we are able to apply well-studied Markov theory such as ergodicity to the forward process. We begin with the definition of Markov transition kernels as in (pp. 151, Le Gall [25]).

Definition 3.3 (Markov transition kernel)**.** *Let $(E, \mathcal{E})$ be a measurable space. A Markov transition kernel is a mapping $K\colon \mathcal{E} \times E \to [0, 1]$ that satisfies the following two properties.*

(1) For every element $x \in E$, the mapping $K(\cdot, x)\colon \mathcal{E} \to [0, 1]$ defines a probability measure on $(E, \mathcal{E})$. We also use the notation $K(\mathrm{d}y, x)$ to refer to the probability measure.

(2) For every set $A \in \mathcal{E}$, the mapping $K(A, \cdot)\colon E \to [0, 1]$ is Borel-$\mathcal{E}$-measurable.

We consider the measurable space $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, whereby $\mathcal{B}(\mathbb{R}^d)$ denotes the Borel σ-algebra on $\mathbb{R}^d$. This allows us to define a sequence of Markov transition kernels used for our forward process. In the following, we fix the length of the forward process as $T \in \mathbb{N}$.

Definition 3.4 (Gaussian diffusion kernels)**.** *Consider the measurable space $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ and a strictly positive real-valued sequence $(\beta_t)_{t=1}^T$ with $\beta_t \in (0, 1)$ for all $t \in \{1, \dots, T\}$ called the variance schedule. Then the functions $K_1, \dots, K_T$ defined by*

$$K_t(dy, x) := \mathcal{N}(\sqrt{1 - \beta_t}x, \beta_t \mathrm{I}_d)$$

for $t \in \{1, \dots, T\}$ are called Gaussian diffusion kernels. In the definition, we refer to the d-dimensional Gaussian distribution and I_d is the $d \times d$ identity matrix.

These are in fact Markovian transition kernels, as property (1) follows from the definition. Property (2) can be shown by Fubini's theorem. Define the integrable and measurable function w.r.t. the product σ-algebra $\mathcal{B}(\mathbb{R}^d) \otimes \mathcal{B}(\mathbb{R}^d)$ as

$$k(y, x) = \frac{1}{(2\pi\beta_t)^{\frac{d}{2}}} e^{-\frac{1}{2\beta_t}(y - \sqrt{1-\beta_t}x)^\top (y - \sqrt{1-\beta_t}x)}$$

and by Fubini's theorem, we have that $x \mapsto \int_A k(y, x)\mathrm{d}y = K_t(A, x)$ is measurable. Using the diffusion kernels, we can define the forward process as a time-inhomogeneous version of the Markov process defined in (pp.152, Le Gall [25]).

Definition 3.5 (Forward process)**.** *Consider a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F})_{t=0,\dots,T}, \mathbb{P})$. Let $Q \in \mathcal{M}^+(\mathbb{R}^d)$ be a probability measure with Lebesgue-density. Furthermore let $(\beta_t)_{t=1}^T$ be a variance schedule and let $K_1, \dots, K_T$ be Gaussian diffusion kernels. Then, an $\mathcal{F}_t$-adapted process $X_0, \dots, X_T$ is called a forward process, if $X_0 \sim Q$ and for all bounded and measurable function $f \in \mathcal{C}_b(\mathbb{R}^d)$, the equality*

$$\mathbb{E}\left[f(X_{t+1}) \mid \mathcal{F}_t\right] = \int_{\mathbb{R}^d} f(y) K_{t+1}(\mathrm{d}y, X_t)$$

holds for $t = 0, \dots, T - 1$.

In the following, we highlight some important observations as a consequence of Definition 3.5. Indeed, the right-hand-side (r.h.s.) is well-defined, as f is assumed to be bounded, so the integral is also always finite. Furthermore the r.h.s. is measurable. For indicator functions of the form $f = \mathbb{1}_A$, we have just shown that the mapping $x \mapsto \int_{\mathbb{R}^d} f(y)K_t(\mathrm{d}y, x)$ is measurable and using measure-theoretic induction, this property can be extended to bounded and measurable functions f. In particular, it is a measurable function of the variable X_t and therefore w.r.t. the sigma-algebra generated by X_t, i.e. $\sigma(X_t)$. As a result, if we consider the natural filtration $\mathcal{F}_t = \sigma(X_s \colon s = 0, \dots, t)$, the intuitive Markov-property can be observed

$$\mathbb{E}\left[f(X_{t+1}) \mid \sigma(X_0, \dots, X_t)\right] = \int_{\mathbb{R}^d} f(y)K_{t+1}(\mathrm{d}y, X_t) = \mathbb{E}\left[f(X_{t+1}) \mid \sigma(X_t)\right],$$

in the sense that the evolution of the forward process does not depend on its history, but only the current state it is in. A further consequence is that the conditional probabilities are given exactly by the diffusion kernels.

$$\mathbb{P}[X_{t+1} \in A \mid X_t] = \mathbb{E}[\mathbb{1}_A(X_{t+1})|\mathcal{F}_t] = K_{t+1}(A, X_t).$$

However, the natural question arises of whether such process actually exists. We will explicitly construct the forward process as done in literature and show that it precisely satisfies Definition 3.5. Later, we will also discuss how the existence of such processes can be guaranteed without explicit constructions, which provides a framework for a larger class of forward processes, one could define.

Consider a random variable $X \sim Q$ and a sequence of i.i.d. standard Gaussian variables $\varepsilon_0, \dots, \varepsilon_{T-1} \sim \mathcal{N}(0, \mathrm{I}_d)$ that are pairwise independent of X as well as a variance schedule $(\beta_t)_{t=1}^T \subset (0, 1)$. Define the forward process as $Z_0 := X$ and

$$Z_t := \sqrt{1 - \beta_t} Z_{t-1} + \sqrt{\beta_t}\varepsilon_{t-1}. \tag{3}$$

for $t = 1, \dots, T$. So in each iteration, we perturb the initial datapoint by some random noise with variance determined by the variance schedule. We equip this process with the natural filtration $\mathcal{F}_t = \sigma(Z_s \colon s = 0, \dots, t)$. Then, for any bounded and measurable function $f \in \mathcal{C}_b(\mathbb{R}^d)$ and $t \in \{0, \dots, T-1\}$, we have

$$\begin{aligned}
\int_{\mathbb{R}^d} f(y)K_{t+1}(\mathrm{d}y, Z_t) &= \int_{\mathbb{R}^d} f(y) \frac{1}{(2\pi\beta_{t+1})^{\frac{d}{2}}} e^{-\frac{1}{2\beta_{t+1}}(y - \sqrt{1-\beta_{t+1}}Z_t)^\top (y - \sqrt{1-\beta_{t+1}}Z_t)} \mathrm{d}y \\
&= \int_{\mathbb{R}^d} f(\sqrt{1 - \beta_{t+1}}Z_t + \sqrt{\beta_{t+1}}y) \frac{1}{(2\pi)^{\frac{d}{2}}} e^{-\frac{1}{2}y^\top y} \mathrm{d}y \\
&= \mathbb{E}_{\varepsilon_t}\left[f(\sqrt{1 - \beta_{t+1}}Z_t + \sqrt{\beta_{t+1}}\varepsilon_t)\right].
\end{aligned} \tag{4}$$

In the second equality, we apply the substitution rule and in the last equality, we take the the expectation only with respect to the standard Gaussian variable $\varepsilon_t \sim \mathcal{N}(0, \mathrm{I}_d)$ denoted by $\mathbb{E}_{\varepsilon_t}[\cdot]$. In order to proceed, we consider the following Corollary.

Corollary 3.6. *Let $(\Omega, \mathcal{A}, \mathbb{P})$ be some probability space and let $X, Y \in \mathcal{L}^1(\mathbb{P})$ be independent $\mathbb{R}^d$-valued random variables. Then, for any bounded and Borel-measurable function $f\colon \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d$, the following equality holds true*

$$\mathbb{E}_Y[f(X,Y)] = \mathbb{E}[f(X,Y) \mid \sigma(X)],$$

where $\mathbb{E}_Y$ denotes the expectation only w.r.t. Y.

Proof. We will use two well-known propositions. First[14] for any $\sigma(X)$-measurable $\mathbb{R}^d$-valued random variable Z, there exists a function $g\colon \mathbb{R}^d \to \mathbb{R}^d$ such that $Z = g(X)$. Secondly[15], we have an alternative characterization of conditional expectations, which states that a $\mathcal{F}$-measurable function $\mathbb{E}[X|\mathcal{F}]$ is the conditional expectation of X given $\mathcal{F}$ if and only if

$$\mathbb{E}[ZX] = \mathbb{E}[Z\mathbb{E}[X|\mathcal{F}]]$$

holds for all $\mathcal{F}$-measurable functions Z with $ZX \in \mathcal{L}^1(\mathbb{P})$.

We first show that for bounded and measurable functions $f\colon \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d$, the random variable $\mathbb{E}_Y[f(X,Y)]$ is $\sigma(X)$-measurable. This can be seen as a direct consequence of Fubini/Tonelli's theorem, which states that the restrictions $f_x\colon \mathbb{R}^d \to \mathbb{R}^d, x \mapsto \int_{\mathbb{R}^d} f(x,y)\mathbb{P}(\mathrm{d}y)$ remain measurable. Thus the composition $f_x(X) = \mathbb{E}_Y[f(X,Y)]$ is also $\sigma(X)$-measurable.

Secondly, let Z be a $\sigma(X)$-measurable function with $Zf(X,Y) \in \mathcal{L}^1(\mathbb{P})$ and let $g\colon \mathbb{R}^d \to \mathbb{R}^d$ be a function such that $g(X) = Z$. Then, we have

$$\begin{aligned}
\mathbb{E}\left[g(X)f(X,Y)\right] &= \int_{\mathbb{R}^d\times\mathbb{R}^d} g(x)f(x,y)\mathbb{P}_{(X,Y)}(\mathrm{d}(x,y)) \\
&= \int_{\mathbb{R}^d\times\mathbb{R}^d} g(x)f(x,y)\mathbb{P}_X \otimes \mathbb{P}_Y(\mathrm{d}(x,y)) \\
&= \int_{\mathbb{R}^d} g(x) \int_{\mathbb{R}^d} f(x,y)\mathbb{P}_Y(\mathrm{d}y)\mathbb{P}_X(\mathrm{d}x) \\
&= \int_{\mathbb{R}^d} g(x)\mathbb{E}_Y[f(x,Y)]\mathbb{P}_X(\mathrm{d}x) = \mathbb{E}\left[g(X)\mathbb{E}_Y\left[f(X,Y)\right]\right],
\end{aligned}$$

where we used the independence of X and Y in the second equality and Fubini in the third equality. Using the second proposition, we obtain the desired equality of

$$\mathbb{E}_Y[f(X,Y)] = \mathbb{E}[f(X,Y) \mid \sigma(X)].$$

□

Returning to verifying that the constructed process is a forward process (4), we apply Corollary 3.6 on the measurable function $\mathbb{R} \times \mathbb{R} \ni (x,y) \mapsto f(\sqrt{1-\beta_{t+1}}x + \sqrt{\beta_{t+1}}y)$ and obtain

$$\begin{aligned}
\mathbb{E}_\varepsilon\left[f(\sqrt{1-\beta_{t+1}}Z_t + \sqrt{\beta_{t+1}}\varepsilon_t)\right] &= \mathbb{E}\left[f(\sqrt{1-\beta_{t+1}}Z_t + \sqrt{\beta_{t+1}}\varepsilon_t) \mid \sigma(Z_t)\right] \\
&= \mathbb{E}\left[f(\sqrt{1-\beta_{t+1}}Z_t + \sqrt{\beta_{t+1}}\varepsilon_t) \mid \mathcal{F}_t\right] \\
&= \mathbb{E}\left[f(Z_{t+1}) \mid \mathcal{F}_t\right].
\end{aligned}$$

[14] Proof is based on standard measure-theoretic induction arguments.

[15] Lemma 1.16 in Prof. Christoph Knochenhauer (Belak) Probability Theory 2 Lecture notes

The second equality can be argued using identical arguments as in Corollary 3.6 by applying the alternative characterization of conditional expectations to $\mathbb{E}\left[f(\sqrt{1-\beta_{t+1}}Z_t + \sqrt{\beta_{t+1}}\varepsilon_t) \mid \mathcal{F}_t\right]$. Writing g as a function of $Z_0, \dots, Z_t$ and using the independence of ε_t with $Z_0, \dots, Z_t$ shows that the conditional expectations matches the left hand side (see Appendix for details). This finally proves that the constructed process $Z_0, \dots, Z_T$ does in fact satisfy Definition 3.5.

Next, we want to show that the Definition of a forward process implies the existence of a joint probability density, which was mentioned but not proven in (pp. 153 Le Gall [25]).

Proposition 3.7. *For a given forward process $X_0, \dots, X_T$ as well as bounded and measurable functions $f_0, f_1 \dots, f_T \in \mathcal{C}_b(\mathbb{R}^d)$, we have*

$$\begin{aligned}&\mathbb{E}[f_0(X_0)f_1(X_1)\cdots f_T(X_T)]\\ &= \int_{\mathbb{R}^d}\int_{\mathbb{R}^d}\cdots\int_{\mathbb{R}^d} f(x_0)f(x_1)\cdots f(x_T)K_T(\mathrm{d}x_T, x_{T-1})\cdots K_1(\mathrm{d}x_1, x_0)\mathrm{d}Q(x_0),\end{aligned}$$

Proof. This can be shown using induction on T. For $T = 1$, we have by Definition 3.5

$$\begin{aligned}\mathbb{E}[f_0(X_0)f_1(X_1)] &= \mathbb{E}\left[\mathbb{E}[f_0(X_0)f_1(X_1) \mid \mathcal{F}_0]\right]\\ &= \mathbb{E}\left[f_0(X_0)\mathbb{E}[f_1(X_1 \mid \mathcal{F}_0]\right]\\ &= \mathbb{E}\left[f_0(X_0)\int_{\mathbb{R}^d} f_1(x_1)K_1(\mathrm{d}x_1, X_0)\right]\\ &= \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} f_0(x_0)f_1(x_1)K_1(\mathrm{d}x_1, x_0)\mathrm{d}Q(x_0).\end{aligned}$$

Assume the equality for $T = n \in \mathbb{N}$ for the induction step. Then using similar arguments for $n+1$, we have

$$\begin{aligned}&\mathbb{E}[f_0(X_0)f_1(X_1)\cdots f_{n+1}(X_{n+1})]\\ &= \mathbb{E}\left[\mathbb{E}[f_0(X_0)f_1(X_1)\cdots f_{n+1}(X_{n+1}) \mid \mathcal{F}_n]\right]\\ &= \mathbb{E}\left[f_0(X_0)f_1(X_1)\cdots f_n(X_n)\int_{\mathbb{R}^d} f_{n+1}(x_{n+1})K_{n+1}(\mathrm{d}x_{n+1}, X_n)\right]\\ &= \int_{\mathbb{R}^d}\int_{\mathbb{R}^d}\cdots\int_{\mathbb{R}^d} f_0(x_0)f_1(x_1)\cdots f_{n+1}(x_{n+1})K_{n+1}(\mathrm{d}x_{n+1}, x_{T-1})\cdots K_1(\mathrm{d}x_1, x_0)\mathrm{d}Q(x_0),\end{aligned}$$

where in the last equality we used the induction assumption on the function $\tilde{f}(X_n) := f_n(X_n)\int_{\mathbb{R}^d} f_{n+1}(x_{n+1})K_{n+1}(\mathrm{d}x_{n+1}, X_n)$, which remains bounded and measurable as a product of bounded and measurable functions. □

Using monotone-class arguments, it can be shown that for any process that satisfies Proposition 3.7, also defines a forward process. Then one can show the existence of this process by defining the joint distribution and using Kolmogorov's extension theorem. Note that the choice of Gaussian transition kernels was somewhat arbitrary and one may also define other transition kernels. In those cases, the existence can be argued by checking for regularity and Kolmogorov's extension theorem.

A consequence of Proposition 3.7 is choosing $f_1 := \mathbb{1}_{A_1}, \dots, f_T := \mathbb{1}_{A_T}$ for some $A_1, \dots, A_T \in \mathcal{B}(\mathbb{R}^d)$, which implies that the joint probability distribution is given by

$$\begin{aligned}\mathbb{P}[X_0 \in A_0, X_1 \in A_1, \dots, X_T \in A_T] &= \mathbb{E}[\mathbb{1}_{A_0}(X_0)\mathbb{1}_{A_1}(X_1)\cdots\mathbb{1}_{A_T}(X_T)] \\ &= \int_{A_0}\int_{A_1}\cdots\int_{A_T} K_T(\mathrm{d}x_T, x_{T-1})\cdots K_1(\mathrm{d}x_1, x_0)\mathrm{d}Q(x_0).\end{aligned} \tag{5}$$

Notice that by assumption all of the probability measures have a Lebesgue-density and so the joint density is simply given by the product of the densities. As discussed in the motivation, a joint probability density allows us to define conditional densities. Throughout this thesis, we will use the following notation. For $s, t, r, u \in \{0, \dots, T\}$, let $q_{X_t}(\cdot)$ be the density of X_t and we set the conditional densities as

$$q(x_t \mid x_s) := \frac{q_{(X_t, X_s)}(x_t, x_s)}{q_{X_s}(x_s)}$$

and

$$q(x_{s:t} \mid x_{r:u}) := \frac{q_{(X_s,\dots,X_t,X_r,\dots,X_u)}(x_s, \dots, x_t, x_r, \dots, x_u)}{q_{(X_r,\dots,X_u)}(x_r, \dots, x_u)}.$$

Note that we should additionally require that the density $q(x)$ of the data distribution Q is strictly positive, which ensures that all conditional probabilities are well-defined. Furthermore, it is easily verified that these conditional densities preserve the Markovian property in the sense that for $s < t$, we have

$$q(x_t \mid x_s) = q(x_t \mid x_{0:s}). \tag{6}$$

In practice, we do not have to sample T standard Gaussian variables to compute the forward process, as the following proposition shows that it is sufficient to sample one standard Gaussian variable and obtain X_t immediately from X_0. From this point on, denote the density of the multivariate normal distribution with mean μ and covariance-matrix Σ as $f_{\mathcal{N}}(x \mid \mu, \Sigma)$.

Proposition 3.8. *Let $X_0, \dots, X_T$ be a forward process with variance schedule $(\beta_t)_{t=1}^T$. Define $\alpha_t := 1 - \beta_t$ for $t = 1, \dots, T$ and $\bar{\alpha}_t := \prod_{i=1}^t \alpha_i$, then, if we condition on X_0, the conditional probability densities are given by*

$$q(x_t \mid x_0) = f_{\mathcal{N}}(x_t \mid \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)\mathrm{I}_d).$$

Proof. We will show this by strong induction. For $t = 1$, it is obvious by equation (5). For $t = 2$, we use the explicit construction of the forward process as in (3) and obtain

$$\begin{aligned}X_2 &= \sqrt{1-\beta_2}\left(\sqrt{1-\beta_1}X_0 + \sqrt{\beta_1}\varepsilon_0\right) + \sqrt{\beta_2}\varepsilon_1 \\ &= \sqrt{\bar{\alpha}_2}X_0 + \sqrt{\beta_1 - \beta_1\beta_2}\varepsilon_0 + \sqrt{\beta_2}\varepsilon_1 \\ &\overset{d}{=} \sqrt{\bar{\alpha}_2}X_0 + \sqrt{(1-\bar{\alpha}_2)}\varepsilon,\end{aligned}$$

where the last equality is understood in the sense of distribution and $\varepsilon \sim \mathcal{N}(0, \mathrm{I}_d)$. If we assume that the equality holds for all $s \leqslant t \in \mathbb{N}$, it follows that

$$\begin{aligned} q(x_{t+1} \mid x_0) &= \int q(x_{t+1} \mid x_t) q(x_t \mid x_0) \mathrm{d}x_t \\ &= \int_{\mathbb{R}^d} f_{\mathcal{N}}(x_{t+1} \mid \sqrt{(1-\beta_{t+1})} x_t, \beta_{t+1} \mathrm{I}_d) f_{\mathcal{N}}(x_t \mid \sqrt{\bar{\alpha}_t} x_0, (1-\bar{\alpha}_t) \mathrm{I}_d) \mathrm{d}x_t \\ &= f_{\mathcal{N}}(x_{t+1} \mid \sqrt{\bar{\alpha}_{t+1}} x_0, (1-\bar{\alpha}_{t+1}) \mathrm{I}_d). \end{aligned}$$

where the first equality can be derived from representation (5). The last equality can be seen as the induction hypothesis for the case $t = 2$ applied to a new forward process with variance schedule $\beta_1 := 1 - \bar{\alpha}_t$ and $\beta_2 := \beta_{t+1}$. □

Applying the previous Proposition to the case where $t = T$, yields

$$q(x_T \mid x_0) = f_{\mathcal{N}}(x_T \mid \sqrt{\bar{\alpha}_T} x_0, (1-\bar{\alpha}_T) \mathrm{I}_d).$$

This shows that the forward process for a given data-point X_0 up to time t can be computed immediately in practice by sampling one standard Gaussian variable $\varepsilon \sim \mathcal{N}(0, \mathrm{I}_d)$ and computing $X_t = \sqrt{\bar{\alpha}_t} X_0 + \sqrt{(1-\bar{\alpha}_t)} \varepsilon$.

An important assumption for the forward process is that $q(x_T) \approx \mathcal{N}(0, \mathrm{I}_d)$. In the following, we will show that the forward process converges in distribution against $\mathcal{N}(0, \mathrm{I}_d)$ as $T \to \infty$. This also highlights the advantage of our built-up Markov framework, as we are able to apply the theory of ergodicity. In the context of the forward process, we are going to modify the theory slightly from Feller [16], which only considered real-valued time-homogeneous Markov chains that depend on one transition kernel. First, note that for some initial distribution Q, the marginal distributions of the forward process after n-steps are recursively given by

$$\begin{aligned} Q_1(A) &:= \int_{\mathbb{R}^d} K_1(A, x_0) \mathrm{d}Q(x_0) \\ Q_n(A) &:= \int_{\mathbb{R}^d} K_n(A, x_{n-1}) \mathrm{d}Q_{n-1}(x_{n-1}), \end{aligned} \tag{7}$$

which follows from the joint distribution (5).

Definition 3.9 (Stationary distribution[16])**.** *For a given forward process $X_0, \ldots, X_T$, let α_0 be the initial distribution, i.e. $X_0 \sim \alpha_0$. Then, α_0 is called stationary, if*

$$\alpha_0 = \alpha_1 = \cdots = \alpha_T$$

holds true, where α_t is defined in the sense of (7) for $t = 1, \ldots, T$.

Before introduction the notion of ergodicity for kernels, we require the definition of strictly positive measures, which are both found in pp. 271 of Feller [16].

[16] William Feller, *An Introduction to Probability Theory and Its Applications*, vol. 2 (New York: John Wiley & Sons, 1966), pp. 207.

Definition 3.10 (Strictly positive measures)**.** *A probability measure α is called strictly positive on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, if $\alpha(A) > 0$ holds for all sets of the form $A := [a_1, b_1] \times \cdots \times [a_d, b_d] \in \mathcal{B}(\mathbb{R}^d)$ with $a_i < b_i$ for $i = 1, \ldots, d$. Analogously, a Markovian transition kernel K is strictly positive, if $K(A, x) > 0$ holds for all $x \in \mathbb{R}^d$ and previously defined $A \in \mathcal{B}(\mathbb{R}^d)$.*

Definition 3.11 (Ergodic Markov Kernels)**.** *A sequence of Markovian transition kernels $(K_n)_{n\in\mathbb{N}}$ are ergodic, if there exists a strictly positive probability distribution α such that for any initial distribution γ_0, we have $\gamma_n \xrightarrow{d} \alpha$ for $n \longrightarrow \infty$ in the sense of (7).*

To ensure ergodicity, we need to impose certain regularity conditions on the Markov kernels, so the paths of the forward process converge to some distribution. This can be achieved by considering continuous and bounded functions $u_0 \in \mathcal{C}_b(\mathbb{R}^d)$ and recursively defining

$$u_n(x) = \int_{\mathbb{R}^d} u_{n-1}(y) \mathrm{d}K_n(\mathrm{d}y, x) \tag{8}$$

for $n \geqslant 1$. From this we obtain a family of transformations, where regularity can be imposed.

Definition 3.12 (Regular kernel[17])**.** *A sequence of Markovian transition kernel $(K_n)_{n\in\mathbb{N}}$ are called regular on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)$ if the family of transformations $(u_n)_{n\in\mathbb{N}} \subset \mathcal{C}_b(\mathbb{R}^d)$ as in (8), is equicontinuous whenever u_0 is uniformly continuous on $\mathbb{R}^d$. Equicontinuity is understood in the sense that for every $\varepsilon > 0$, there exists a $\delta > 0$, such that*

$$|x - y| < \delta \implies |u_k(x) - u_k(y)| < \varepsilon$$

holds true for all $x, y \in \mathbb{R}^d$ and $k \in \mathbb{N}$.

Using this regularity constraint, it is possible to determine when a strictly positive kernel is ergodic and what the limiting distribution is.

Theorem 3.13. *A sequence of strictly positive and regular transition kernels $(K_n)_{n\in\mathbb{N}}$ are ergodic if there exists a strictly positive stationary distribution α.*

A proof of this theorem for a transition kernel of a time-homogeneous and real-valued Markov chain can be found on page 273 (Feller [16]). This proof works nearly identically in our case, but we need the additional assumption that there exists some $N \in \mathbb{N}$ such that $K_n = K_m$ for $n, m \geqslant N$, which we also assume in the following. The proof is provided in the Appendix.

Proposition 3.14. *Let $(K_n)_{n\in\mathbb{N}}$ be Gaussian diffusion kernels such that there exists some $N \in \mathbb{N}$ such that $K_n = K_m$ for $n, m \geqslant N$. Then the kernels are ergodic with strictly positive stationary distribution $\alpha_0 := \mathcal{N}(0, \mathrm{I}_d)$. In other words for any initial distribution γ_0, we have $\gamma_T \to \mathcal{N}(0, \mathrm{I}_d)$ for $T \to \infty$ in the sense of (7).*

[17] William Feller, *An Introduction to Probability Theory and Its Applications*, vol. 2 (New York: John Wiley & Sons, 1966), pp. 207.

Proof. Clearly the Gaussian diffusion kernels $(K_n)_{n\in\mathbb{N}}$ and α_0 are strictly positive. In order to show regularity, we apply the substitution rule $z = y - \sqrt{1-\beta_n}x$

$$\begin{aligned}
u_n(x) &= \int_{\mathbb{R}^d} u_{n-1}(y) K_n(\mathrm{d}y, x) \\
&= \int_{\mathbb{R}^d} u_{n-1}(y) f_{\mathcal{N}}(y \mid \sqrt{1-\beta_n}x, \beta_n \mathrm{I}_d)\mathrm{d}y \\
&= \int_{\mathbb{R}^d} u_{n-1}\left(z + \sqrt{1-\beta_n}x\right) f_{\mathcal{N}}\left(z + \sqrt{1-\beta_n}x \mid \sqrt{1-\beta_n}x, \beta_n \mathrm{I}_d\right) \mathrm{d}z \\
&= \int_{\mathbb{R}^d} u_{n-1}\left(z + \sqrt{1-\beta_n}x\right) f_{\mathcal{N}}\left(z \mid 0, \beta_n \mathrm{I}_d\right) \mathrm{d}z.
\end{aligned}$$

We will now show by induction that $(u_n)_{n\in\mathbb{N}}$ is equicontinuous using the same choice of $\delta > 0$. Let $\varepsilon > 0$ be arbitrary, then by assumption u_0 is uniformly continuous, so there exists $\delta > 0$ such that for $x, y \in \mathbb{R}^d$ with $|x - y| < \delta$, we have $|u_0(x) - u_0(y)| < \varepsilon$. For the induction step, assume the uniform continuity of u_n, then for $x, y \in \mathbb{R}^d$ with $|x - y| < \delta$, the following relation holds

$$\begin{aligned}
|u_{n+1}(x) - u_{n+1}(y)| &= \left| \int_{\mathbb{R}^d} \left(u_n(z + \sqrt{1-\beta_n}x) - u_n(z + \sqrt{1-\beta_n}y) \right) f_{\mathcal{N}}\left(z \mid 0, \beta_n \mathrm{I}_d\right) \mathrm{d}z \right| \\
&\leqslant \int_{\mathbb{R}^d} \left| u_n(z + \sqrt{1-\beta_n}x) - u_n(z + \sqrt{1-\beta_n}y) \right| f_{\mathcal{N}}\left(z \mid 0, \beta_n \mathrm{I}_d\right) \mathrm{d}z \\
&\leqslant \varepsilon \int_{\mathbb{R}^d} f_{\mathcal{N}}\left(z \mid 0, \beta_n \mathrm{I}_d\right) \mathrm{d}z = \varepsilon,
\end{aligned}$$

whereby the last inequality follows from the induction hypothesis for u_n, as $|z + \sqrt{1-\beta_n}x - \left(z + \sqrt{1-\beta_n}y\right)| = |\sqrt{1-\beta_n}(x-y)| < \delta$ holds, since $\beta_n \in (0,1)$.

Next, we want to show that α_0 is a stationary distribution. We calculate

$$\begin{aligned}
\alpha_1(A) &= \int_{\mathbb{R}^d} K_1(A, x_0)\alpha_0(\mathrm{d}x_0) \\
&= \int_{\mathbb{R}^d} \int_A f_{\mathcal{N}}(x_1 \mid \sqrt{1-\beta_1}x_0, \beta_1 \mathrm{I}_d)\mathrm{d}x_1 f_{\mathcal{N}}(x_0 \mid 0, \mathrm{I}_d)\mathrm{d}x_0 \\
&= \int_A \int_{\mathbb{R}^d} f_{\mathcal{N}}(x_1 \mid \sqrt{1-\beta_1}x_0, \beta_1 \mathrm{I}_d) f_{\mathcal{N}}(x_0 \mid 0, \mathrm{I}_d)\mathrm{d}x_0 \mathrm{d}x_1 \qquad (9) \\
&= \int_A f_{\mathcal{N}}(x_1 \mid 0, \mathrm{I}_d)\mathrm{d}x_1, \qquad (10)
\end{aligned}$$

where we use Fubini in (9) as well as the same argument as in the proof of Proposition 3.8 in (10). Clearly this equality also holds for α_k for $k \geqslant 1$ and by Theorem 3.13, we conclude that the Gaussian diffusion kernels are ergodic with stationary distribution α_0. □

An interesting question is how to choose the variance schedule. If the variance-schedule is close to 1, one would expect faster convergence to its stationary distribution $\mathcal{N}(0, \mathrm{I}_d)$, however, this does not translate to improvement in empirical performance. In practice, we

only consider a finite amount of denoising steps and want to ensure that $p_\theta(x_T) \approx \mathcal{N}(0, \mathrm{I}_d)$. In the original DDPM paper, Ho et al. [20] choose $T = 1000$ as well as a linear variance schedule from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$, meaning the added noise increases linearly in the time steps. On the other hand, Nichol and Dhariwal [29] suggest improvements for the variance schedule, as they claim that a linear schedule is too noisy towards the end, which does not improve sample quality. They suggest $T = 3000$ with a cosine variance schedule, defined in terms of $\bar{\alpha}_t$ (Proposition 3.8), where

$$\bar{\alpha}_t = \frac{g(t)}{g(0)}, \quad g(t) := \cos\left(\frac{t/T + s}{1+s} \cdot \frac{\pi}{2}\right)^2$$

and found improved results on the image benchmark datasets ImageNet and CIFAR. Yet, it is also possible to learn the variance schedule, as Kingma et al. [24] use a monotonic neural network to learn the variance schedule.

Deriving Backward Process

We now want to learn the denoising/backward process or the conditionals $q(x_{t-1} \mid x_t)$. As explained in the guideline, if we are able to model these conditional probabilities $p_\theta(x_{t-1} \mid x_t)$, then we are also able to sample from the initial distribution using

$$p_\theta(x_{0:T}) = p_\theta(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t)$$

and the assumption that $p_\theta(x_T) = \mathcal{N}(0, \mathrm{I}_d)$. Note that this equality holds true, as the reverse forward process $X_T, \ldots, X_0$ remains a Markov process and therefore exhibits the Markov property. This can be easily verified by rearranging the explicit construction of the forward process in (3), i.e.

$$Z_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \left(Z_t - \sqrt{\beta_t}\varepsilon_{t-1}\right).$$

For this, we return to the aim of minimizing the KL-divergence of P_θ and Q or equivalently the minimization of the negative log-likelihood. The following proposition shows an upper bound of the negative log-likelihood using the forward process that turns out to be tractable.

Proposition 3.15. *For a given forward process $X_0, \ldots, X_T$, we have*

$$\mathbb{E}_{q(x_0)}\left[-\log p_\theta(x_0)\right] \leqslant -\mathbb{E}_{q(x_0, x_{1:T})}\left[\log \frac{p_\theta(x_0, x_{1:T})}{q(x_{1:T} \mid x_0)}\right]$$

Proof. A sequence of straightforward equations shows

$$\begin{aligned}
\mathbb{E}_{q(x_0)}\left[-\log p_\theta(x_0)\right] &= \mathbb{E}_{q(x_0,x_{1:T})}[-\log p_\theta(x_0)] \\
&= \mathbb{E}_{q(x_0,x_{1:T})}\left[\log \frac{p_\theta(x_{1:T} \mid x_0)}{p_\theta(x_0, x_{1:T})}\right] \\
&= \mathbb{E}_{q(x_0,x_{1:T})}\left[\log \frac{q(x_{1:T} \mid x_0)p_\theta(x_{1:T} \mid x_0)}{p_\theta(x_0, x_{1:T})q(x_{1:T} \mid x_0)}\right] \\
&= \mathbb{E}_{q(x_0,x_{1:T})}\left[\log \frac{q(x_{1:T} \mid x_0)}{p_\theta(x_0, x_{1:T})}\right] - \mathbb{E}_{q(x_0,x_{1:T})}\left[-\log \frac{p_\theta(x_{1:T} \mid x_0)}{q(x_{1:T} \mid x_0)}\right] \\
&= -\mathbb{E}_{q(x_0,x_{1:T})}\left[\log \frac{p_\theta(x_0, x_{1:T})}{q(x_{1:T} \mid x_0)}\right] - \mathbb{E}_{q(x_0)}\left[\mathbf{d}_{\mathrm{KL}}\left(q(x_{1:T} \mid x_0) \| p_\theta(x_{1:T} \mid x_0)\right)\right] \\
&\leqslant -\mathbb{E}_{q(x_0,x_{1:T})}\left[\log \frac{p_\theta(x_0, x_{1:T})}{q(x_{1:T} \mid x_0)}\right],
\end{aligned}$$

whereby the inequality follows from the non-negativity of the KL-divergence. □

It turns out this bound is computable and is a slight modification of the so-called evidence lower bound (ELBO). However, the term $\mathbb{E}_{q(x_0)}\left[\mathbf{d}_{\mathrm{KL}}\left(q(x_{1:T} \mid x_0) \| p_\theta(x_{1:T} \mid x_0)\right)\right]$ is neglected in the inequality, which depends on the model distribution p_θ. Therefore, this bound may not be tight and minimizing the upper bound may not minimize the negative log-likelihood. We informally describe why this bound is still meaningful by considering a wider range of model called variational Auto-encoders shows.

At a high level, these model assume that the datapoints $x \sim q(x)$ are based on some underlying latent variable z. The so-called encoder attempts to model the conditional probabilities $q(z \mid x)$ using some parametrized model $p_\theta(z \mid x)$, by minimizing the KL-divergence $\mathbf{d}_{\mathrm{KL}}(p_\theta(z \mid x) \| q(z \mid x))$. Then a similar set of calculations as in the proof of the previous Proposition detailed in the Appendix, shows

$$\min_{\theta \in \Theta} \mathbb{E}_{q(x)}\left[\mathbf{d}_{\mathrm{KL}}\left(q(z \mid x) \| p_\theta(z \mid x)\right)\right] = \min_{\theta \in \Theta} -\mathbb{E}_{q(x)p_\theta(z|x)}\left[\log \frac{p_\theta(x, z)}{q(z \mid x)}\right].$$

If we assume the left-hand-side is small, we can expect $p_\theta(z \mid x) \approx q(z \mid x)$ and

$$\mathbb{E}_{q(x)p_\theta(z|x)}\left[\log \frac{p_\theta(x, z)}{q(z \mid x)}\right] \approx \mathbb{E}_{q(x,z)}\left[\log \frac{p_\theta(x, z)}{q(z \mid x)}\right].$$

Notice that the negative r.h.s. is exactly the bound discussed in Proposition 3.15, and diffusion models can then be seen as the case where the latent variables z is the forward process $x_{1:T}$. Minimizing the objective on the right-hand-side also suggests that the neglected term remains small. We will now show that the bound from Proposition 3.15 is computable and derive the backward process of diffusion models. The proofs of these equalities are well-known in literature and based on pure calculation and do not provide additional insight, but we will include them in the Appendix or reference them for completeness.

Proposition 3.16. *For the upper bound from Proposition 3.15, the following equality holds*

$$
\begin{aligned}
-\mathbb{E}_{q(x_{0:T})}\left[\log \frac{p_\theta(x_0, x_{1:T})}{q(x_{1:T} \mid x_0)}\right] &= \mathbb{E}_{q(x_{0:T})}\left[-\log p_\theta\left(x_0|x_1\right)\right] \\
&\quad + \sum_{t=2}^{T} \mathbb{E}_{q(x_0,x_t)}\left[\mathbf{d}_{\mathrm{KL}}\left(q(x_{t-1} \mid x_0, x_t) \| p_\theta(x_{t-1}|x_t)\right)\right] \\
&\quad + \mathbb{E}_{q(x_0,x_T)}\left[\mathbf{d}_{\mathrm{KL}}\left(q(x_T|x_0) \| p_\theta(x_T)\right)\right]
\end{aligned}
$$

Proof. See Appendix and can also be found on page 12 of [26]. □

If we take a look at the terms from Proposition 3.16 a bit closer, we are able to identify three main components. Luo [26] explains these as follows:

(1) Reconstruction term $\mathbb{E}_{q(x_0,x_{1:T})}\left[-\log p_\theta\left(x_0|x_1\right)\right]$: measures how well the model $p_\theta\left(x_0 \mid x_1\right)$, is able to reconstruct the data points x_0 after one step in the diffusion process x_1.

(2) Denoising-matching term $\sum_{t=2}^{T} \mathbb{E}_{q(x_0,x_t)}\left[\mathbf{d}_{\mathrm{KL}}\left(q(x_{t-1} \mid x_0, x_t) \| p_\theta(x_{t-1}|x_t)\right)\right]$: is a measure of how well the model matches the backwards/denoising steps $p_\theta(x_{t-1} \mid x_t)$ to the actual and tractable denoising steps $q(x_{t-1}|x_t, x_0)$.

(3) Prior-matching term: $\mathbb{E}_{q(x_0,x_T)}\left[\mathbf{d}_{\mathrm{KL}}\left(q(x_T|x_0) \| p_\theta(x_T)\right)\right]$: indicates the extent of which the distribution in the final step of the forward process $q(x_T|x_0)$ matches $p_\theta(x_T)$, which is assumed to be standard Gaussian $\mathcal{N}(0, \mathrm{I}_d)$.

Notice that for term (3) the assumption $p_\theta(x_T) = \mathcal{N}(0, \mathrm{I}_d)$ implies that it is independent of the model parameter θ and therefore does not require optimization, if T is fixed. However, we neglect term (1), as most of the cost lies in the denoising-matching term (2). Thus, we are finally able to define the optimization objective L as the denoising-matching term (2)

$$
L(\theta) := \sum_{t=2}^{T} \mathbb{E}_{q(x_0,x_t)}\left[\mathbf{d}_{\mathrm{KL}}\left(q(x_{t-1} \mid x_0, x_t) \| p_\theta(x_{t-1}|x_t)\right)\right] \tag{11}
$$

In the following, we will show that this objective is computable. The following Corollary shows that $q(x_{t-1}|x_t, x_0)$ is in fact tractable.

Corollary 3.17. *Define $\alpha_t := 1 - \beta_t$ and $\bar{\alpha}_t := \prod_{i=1}^{t} \alpha_i$ as in Proposition 3.8. Then, the conditional densities $q(x_{t-1} \mid x_t, x_0)$ are Gaussian with*

$$
q(x_{t-1} \mid x_t, x_0) = f_{\mathcal{N}}\left(x_{t-1}; \frac{\sqrt{\alpha_t}\left(1-\bar{\alpha}_{t-1}\right) x_t + \left(1-\alpha_t\right)\sqrt{\alpha_{t-1}} x_0}{1-\bar{\alpha}_t}, \frac{\left(1-\alpha_t\right)\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}\mathrm{I}_d\right).
$$

Proof. The proof relies on Bayes formula as well as long calculations and can be found on page 12 of [26]. □

In the following, we set the true mean and variance from the previous Corollary as

$$\mu_q(x_t, x_0) := \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})\,x_t + (1-\alpha_t)\sqrt{\alpha_{t-1}}x_0}{1-\bar{\alpha}_t}$$

and

$$\sigma_q^2(t) := \frac{(1-\alpha_t)\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}.$$

In effort to minimize the objective L (11), a natural choice of the model distribution $p_\theta(x_{t-1} \mid x_t)$ in light of Corollary 3.17, would be Gaussian, i.e.

$$p_\theta(x_{t-1} \mid x_t) = f_{\mathcal{N}}\left(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)\right) \tag{12}$$

and we only have to model the mean $\mu_\theta(x_t, t)$ and covariance matrix $\Sigma_\theta(x_t, t)$. Notice that the difference of the model distribution $p_\theta(x_{t-1} \mid x_t)$ to the actual distribution $q(x_{t-1} \mid x_t, x_0)$ is that we do not know the initial data-point x_0. However, the covariance matrix in Corollary 3.17 does not depend on x_0, so we can choose

$$\Sigma_\theta(x_t, t) := \sigma_q^2(t)\mathrm{I}_d. \tag{13}$$

However, for $\mu_\theta(x_t, t)$ we cannot simply set this as $\mu_q(x_t, x_0)$, since x_0 is unknown. Nevertheless, with the help of the explicit construction of the forward process (3), the following Corollary provides an alternative characterization to $\mu_q(x_t, x_0)$.

Corollary 3.18. *Using the explicit construction of the forward process as in (3), i.e.* $X_t = \sqrt{\bar{\alpha}_t}X_0 + (1-\bar{\alpha}_t)\varepsilon_t$ *with* $\varepsilon_t \sim \mathcal{N}(0, \mathrm{I}_d)$. *Then, the following equality holds true*

$$\mu_q(x_t, x_0) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_t\right)$$

Proof. The proof can be found on page 15 of [26]. □

Using Corollary 3.18, we are inclined to choose the model mean $\mu_\theta(x_t, t)$ as

$$\mu_\theta(x_t, t) := \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta(x_t, t)\right) \tag{14}$$

whereby $\varepsilon_\theta(x_t, t)$ is a parameterized neural network that attempts to predict the noise ε_t. One may ask themselves why we did not model μ_q directly or the initial data-point x_0 in the definition μ_q. This is certainly possible and discussed in [26], yet modelling the noise achieved better empirical results.

It turns out that the KL-divergence between two Gaussian distributions is given in a closed form. And given some assumptions, we are finally able to determine a tractable objective L.

Proposition 3.19. *Assume that the model distribution $p_\theta(x_{t-1} \mid x_t)$ is Gaussian (12) with model covariance matrix (13) and model mean (14), then for the optimization objective L defined in (11), the following equality holds true*

$$L(\theta) = \sum_{t=2}^{T} \frac{1}{2\sigma_q^2(t)} \frac{(1-\alpha_t)^2}{\alpha_t (1-\bar{\alpha}_t)} \cdot \mathbb{E}\left[\|\varepsilon_t - \varepsilon_\theta(X_t, t)\|_2^2\right].$$

Proof. See Appendix and can also be found on page 15 of [26]. □

Ho et al. [20] were able to achieve better empirical results by optimizing the objective L without the weighted term before the expectation. They therefore suggest to use the simple optimization objective.

$$L^{\text{simple}} = \sum_{t=2}^{T} \mathbb{E}\left[\|\varepsilon_t - \varepsilon_\theta(X_t, t)\|^2\right]. \tag{15}$$

To learn the noise estimator ε_θ from a dataset, we sample $x^{(0)}$ from the dataset as well as t uniformly from $\{1, \dots, T\}$. Then we compute $x^{(t)}$, i.e. the forward process of $x^{(0)}$ up to t and evaluate a noise estimate $\varepsilon_\theta(x^{(t)}, t)$ and perform gradient descent on the noise estimate. The pseudo-code in Algorithm 1 shows the training process using this simple objective.

Algorithm 1 Diffusion model training with learning rate α and objective L^{simple}

Require: Dataset $\{x_i^{(0)}\}_{i \in \mathbb{N}}$ with $x_i^{(0)} \sim Q$

1: **repeat**:
2: Consider data-point $x_i^{(0)}$
3: Initialize $t \sim \text{Uniform}(1, \dots, T)$ and $\varepsilon \sim \mathcal{N}(0, \mathrm{I}_d)$
4: Set $x_i^{(t)} \leftarrow \sqrt{\bar{\alpha}_t} \cdot x_i^{(0)} + \sqrt{1-\bar{\alpha}_t} \cdot \varepsilon$
5: Take gradient step $\theta \leftarrow \theta - \alpha \cdot \nabla_\theta \left\|\varepsilon - \varepsilon_\theta(x_i^{(t)}, t)\right\|_2^2$
6: **until** converged

In terms of the sampling process of the diffusion model, we remember that we have now modelled the conditional probabilities $p_\theta(x_{t-1} \mid x_t) = f_{\mathcal{N}}\left(x_{t-1}; \mu_\theta(x_t), \sigma_q^2(t)\mathrm{I}_d\right)$ and know the joint distribution

$$p_\theta(x_{0:T}) = p_\theta(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t).$$

By assumption $p_\theta(x_T) = \mathcal{N}(0, \mathrm{I}_d)$ holds true, so we begin by sampling $x_T \sim \mathcal{N}(0, \mathrm{I}_d)$. Then we can compute the denoising process from t to $t-1$ for $t = T, \dots, 1$ by sampling $\varepsilon \sim \mathcal{N}(0, \mathrm{I}_d)$

$$\begin{aligned} x_{t-1} &= \mu_\theta(x_t, t) + \sqrt{\sigma_q^2(t)}\varepsilon \\ &= \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta(x_t, t)\right) + \sqrt{\sigma_q^2(t)}\varepsilon. \end{aligned}$$

Algorithm 2 Sampling from diffusion model $x \sim p_\theta(x_0)$

Require: Noise $x_T \sim \mathcal{N}(0, \mathrm{I}_d)$
1: **for** $t = T$ to 1 **do**
2: **if** $t > 1$ **then**
3: $\varepsilon \sim \mathcal{N}(0, \mathrm{I}_d)$
4: **else**
5: $\varepsilon = 0$
6: **end if**
7: $x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \varepsilon_\theta \left(x_t, t \right) \right) + \sqrt{\sigma_q^2(t)} \varepsilon$
8: **end for**
9: **return** x_0

We are also able to extend diffusion models to conditional diffusion models, by taking an additional input c into consideration. Instead of learning the joint distribution $p_\theta(x_{0:T})$, we attempt to learn the conditional distribution $p_\theta(x_{0:T} \mid c)$, i.e.

$$\begin{aligned} p_\theta \left(x_{0:T} \mid c \right) &= p \left(x_T \right) \prod_{t=1}^{T} p_\theta \left(x_{t-1} \mid x_t, c \right) \\ \text{with} \quad p_\theta \left(x_{t-1} \mid x_t, c \right) &= f_{\mathcal{N}} \left(x_{k-1}; \mu_\theta \left(x_t, t, c \right), \sigma_q^2(t) \mathrm{I}_d \right) \\ \text{and} \qquad \mu_\theta \left(x_t, t, c \right) &= \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \varepsilon_\theta(x_t, t, c) \right). \end{aligned} \tag{16}$$

The optimization objective in this case would be

$$L^{\text{simple}} = \sum_{t=2}^{T} \mathbb{E} \left[\left\| \varepsilon_t - \varepsilon_\theta \left(X_t, t, c \right) \right\|^2 \right].$$

3.2.3 Connection to other Approaches

As mentioned, there are other popular approaches in literature to diffusion models. We will briefly illustrate the connection to the SDE approach, which can be seen as a continuous-time extension to the discrete time DDPM.

Stochastic Differential Equations

The use of stochastic differential equations allows for a very natural extension of the discussed forward process to continuous time and was introduced by Song et al. [39] in 2021. This approach defines the variance schedule as a positive function $[0, T] \ni t \mapsto \beta(t)$ and the forward process $(X_t)_{t \in [0,T]}$ as the solution of the following SDE

$$\mathrm{d}X_t = -\frac{1}{2} \beta(t) X_t \mathrm{d}t + \sqrt{\beta(t)} \mathrm{d}B_t, \tag{17}$$

whereby $(B_t)_{t\in[0,T]}$ is a d-dimensional Brownian motion and $X_0 \sim Q$. Indeed, if we discretize the SDE (17) by considering a partition $0 = t_0 < \cdots < t_N = T$ for $N \in \mathbb{N}$ with $\Delta t_i := t_{i+1} - t_i = \frac{T}{N}$ for all $i = 0, \ldots, N-1$. Applying the Euler-Maruyama approximation yields

$$\begin{aligned} X_{t_{i+1}} &\approx X_{t_i} - \frac{1}{2}\beta(t_i)X_{t_i}\Delta t_i + \sqrt{\beta(t_i)}(B_{t_{i+1}} - B_{t_i}) \\ &= \left(1 - \frac{1}{2}\beta(t_i)\Delta t_i\right) X_{t_i} + \sqrt{\beta(t_i)\Delta t_i}\varepsilon \end{aligned}$$

with $\varepsilon \sim \mathcal{N}(0, \mathrm{I}_d)$. Defining $\tilde{\beta}(t) := \frac{N\beta(t)}{T}$ absorbs the Δt_i terms and the first-order Taylor expansion of $\sqrt{1-\tilde{\beta}(t)}$ is given by $1 - \frac{1}{2}\tilde{\beta}(t)$. So for small $\tilde{\beta}$, we have

$$X_{t+1} \approx \sqrt{1-\tilde{\beta}(t_i)}X_{t_i} + \sqrt{\tilde{\beta}(t_i)}\varepsilon,$$

which is exactly the form of the constructed forward process (2) in the discrete-time setting. The SDE (17) can also be solved analytically by applying Itô's product rule on $Y_t := e^{\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s} X_t$

$$\begin{aligned} \mathrm{d}Y_t &= \mathrm{d}e^{\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s} X_t \\ &= \frac{1}{2}\beta(t)e^{\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s} X_t \mathrm{d}t + e^{\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s}\mathrm{d}X_t + \mathrm{d}\langle e^{\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s}, Y_t\rangle. \end{aligned}$$

Since $t \mapsto e^{-\frac{1}{2}\int_0^t \beta(s)}$ is deterministic and continuous, it can easily be verified that the quadratic covariation is 0. We continue the calculation from above by substituting equation (17)

$$\begin{aligned} \mathrm{d}Y_t &= \frac{1}{2}\beta(t)e^{\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s} X_t \mathrm{d}t + e^{\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s}\mathrm{d}X_t + \mathrm{d}\langle e^{\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s}, Y_t\rangle \\ &= \frac{1}{2}\beta(t)e^{\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s} X_t \mathrm{d}t + e^{\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s}\left(-\frac{1}{2}\beta(t)X_t\mathrm{d}t + \sqrt{\beta(t)}\mathrm{d}B_t\right) \\ &= e^{\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s}\sqrt{\beta(t)}\mathrm{d}B_t. \end{aligned}$$

This is equivalent to

$$Y_t = Y_0 + \int_0^t e^{\frac{1}{2}\int_0^u \beta(s)\mathrm{d}s}\sqrt{\beta(u)}\mathrm{d}B_u.$$

Now substituting this back in terms of $X_t = e^{-\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s} Y_t$ and using $X_0 = Y_0$ gives

$$\begin{aligned} X_t &= e^{-\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s}\left(X_0 + \int_0^t e^{\frac{1}{2}\int_0^u \beta(s)\mathrm{d}s}\sqrt{\beta(u)}\mathrm{d}B_u\right) \\ &= e^{-\frac{1}{2}\int_0^t \beta(s)\mathrm{d}s} X_0 + \int_0^t e^{-\frac{1}{2}\int_u^t \beta(s)\mathrm{d}s}\sqrt{\beta(u)}\mathrm{d}B_u. \end{aligned}$$

Next, we want to derive a reverse/denosing process from (17), which is different to the DDPM approach. Under the assumption that X_t exhibits a Lebegue density $p_t(x)$ for all $t \in [0, T]$, Anderson [3] found that it is possible to reverse the SDE (17), i.e. $Y_t = X_{T-t}$, which satisfies

$$\mathrm{d}Y_t = \beta(T-t)\left(-\frac{1}{2}Y_t + \nabla \log p_{T-t}(Y_t)\right)\mathrm{d}t + \sqrt{\beta(T-t)}\mathrm{d}B_t.$$

Then the task becomes to model the gradient of the log probability density $\nabla \log p_t(X_t)$, which is known as the score function. For more details on training, we refer to [39]. Following this, we can again use the Euler-Maruyama method to reverse the forward process and generate new samples.

One advantage of the SDE approach is that the drift coefficient $-\frac{1}{2}\beta(t)$ and the diffusion coefficient $\sqrt{\beta(t)}$ in (17) are variable, which allows for more flexibility for defining a forward process. This was recently demonstrated by de Bortoli [6], who was able to derive a convergence bound for a diffusion model with forward process

$$\mathrm{d}X_t = -\beta(t)X_t\mathrm{d}t + \sqrt{2\beta(t)}\mathrm{d}B_t.$$

The multitude of mathematical tools for diffusion models may also explain their success in recent years, which are used to derive better convergence bounds as well as providing various areas of improvement.

3.3 Limit Order Book Modelling using Diffusion Models

To my current knowledge, diffusion models have not been used in the context of modelling limit order books. Nevertheless, there has been growing interest in the use of diffusion models, when dealing with time series data. In particular, Meijer [27] recently did a survey on diffusion-based architectures for time series forecasting and found that these produced state-out-of-the-art results on numerous datasets. Together with the mentioned disadvantages of GANs compared to diffusion models, it motivates the use of diffusion models for this task.

3.3.1 Problem Formulation

For many applications between mid and high-frequency, it is not necessary to model the continuous order book model, which also requires a very computationally efficient model and substantial data storage space. Furthermore, high-frequency data is particularly noisy and therefore also difficult to model. For this reason, we consider the centered order book from Section 2.3.3 and down-sample this to a lower frequency, which is also known as LOB 'snapshots'.

Definition 3.13 (Centered LOB Snapshot process)**.** *Let $(\bar{O}_t)_{t\in[0,T]}$ be a centered order book process in the sense of Definition 2.12. For $\Delta t > 0$ with $K := \lfloor \frac{T}{\Delta t} \rfloor$, the process $(O_k)_{k=0,\dots,K}$ defined as*

$$O_k := \bar{O}_{k\Delta t}$$

is known as the centered LOB snapshot process with Δt frequency.

For some fixed time $\Delta t > 0$, let $(O_t)_{t=0,\dots,K}$ be a centered LOB snapshot process with frequency Δt. We are interested in learning the distribution of the snapshot O_{t+1} given the information up to time t, which we encode as a state variable $S_t \in \mathbb{R}^s$. In other words,

$$O_{t+1} \sim \mathbb{P}_q(o_{t+1} \mid S_t).$$

The state variable could also include other information like macroeconomic news or temporal data. However, in a Markovian setting the state variable would just be the most recent order book state, i.e. $S_t = O_t$.

For practical and simplicity purposes, we will make some further assumptions. We only include bid queues corresponding to d ticks below the mid-price and the ask queues for $d-1$ ticks above the mid-price. Since the market-clearing operator acts on the order book, the bid/ask queue sizes above/below the mid-price are 0. Furthermore, we simply take the mid-price instead of twice the mid-price. Therefore, our centered order book snapshots take the form $O_t = (X_t, p_t) \in \mathbb{R}^{2d+1}$ with mid-price p_t and queue size

$$X_t = \Bigg(\underbrace{-Q^-_{p_t - d\cdot\delta}(t), \dots, -Q^-_{p_t - 1\cdot\delta}}_{-X^-(t)}, \underbrace{Q^+_{p_t\delta}(t), \dots, Q^+_{p_t + (d-1)\cdot\delta}}_{X^+(t)} \Bigg) \in \mathbb{R}^{2d}. \tag{18}$$

Note that in this representation we round the mid-price up to the nearest tick-size δ and we choose the negative bid queue size to emphasize the difference to the ask-side.

3.3.2 Model Architecture: TimeGrad

The model architecture that we will use is a conditional diffusion model known as TimeGrad and is based on the DDPM approach, which was introduced in 2021 by the Zalando research team [2]. It uses exactly the training and sampling algorithm that were outlined in Algorithm 1 and 2. Since we are dealing with time series data, we have to distinguish between the time index of the data and the forward process. In the following, we denote $O_t^{(n)}$ as the n-th step in the forward process of O_t.

TimeGrad is a conditional diffusion model with noise estimator $\varepsilon_\theta(O_t^{(n)}, n, h_t)$ as in (16) that takes an additional variable h_t. In TimeGrad this is a hidden state, which encodes historical data and other state variables using a recurrent neural network (RNN)

$$h_t = \text{RNN}_\theta(\texttt{concat}(O_{t-1}, c_t), h_{t-1}).$$

The hidden state h_{t-1} also takes additional features C_t as an input. In our case, we only consider the order book state at the previous time O_{t-1}, so we have

$$h_t = \text{RNN}_\theta(O_{t-1}, h_{t-1}). \tag{19}$$

The model architecture for the noise estimator ε_θ is depicted in the following diagram, whereby x_t^n corresponds to $O_t^{(n)}$ in our case

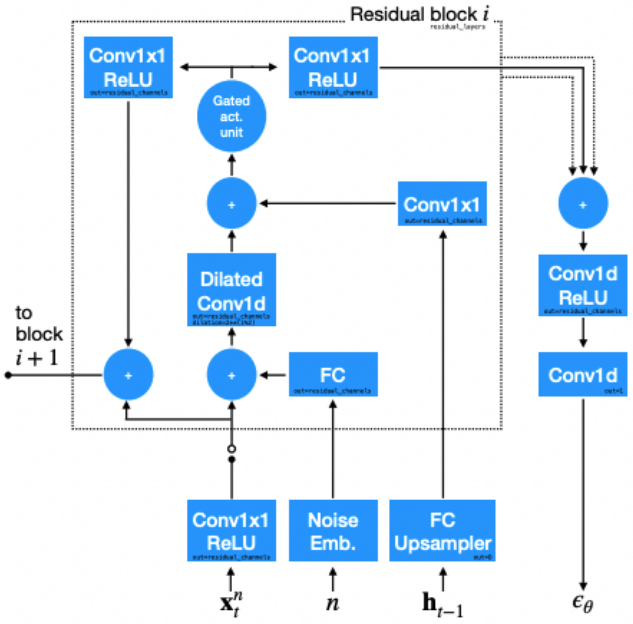


Figure 5: Residual block in TimeGrad Model Architecture [2]

The model consists of 8 so-called residual blocks and the final output ε_θ is a transformation of the sum of each individual residual blocks. Different transformations are also applied to the input, which include embedding and convolutional layers. For more details on the model architecture, we refer to the original paper [2]. An implementation of the TimeGrad can be found in the GluonTS library [17], which we have used.

3.4 Experiment

3.4.1 Dataset

The cryptocurrency exchange Binance offers free Bitcoin/Tether (BTC/USDT) L2 order book data, which we will use in our experiment. The BTC/USDT data was scraped using the Binance API over a ten day period from 10.10.2023 to 19.10.2023[18]. The BTC/USDT orderbook is 100-levels deep, i.e. contains the 100 best ask/bid prices and corresponding bid/ask volumes. The tick-size of BTC/USDT is 0.01 USDT, whilst the mid-price of USDT during the start of the considered period was at 27590.50, which is why we will further aggregate the levels to a tick size of 1 USDT. Furthermore, we choose a 5-second frequency for the centered order book snapshots, i.e. we have transition times $t_1 < \cdots < t_N$ such that $t_{i+1} - t_i = 5$ holds for $i = 1, \ldots N-1$, if we consider time to be measured in seconds. Given these assumptions, it is important to choose a suitable dimension d of the centered order book representation (18). For this, we would ideally like to have

(1) Non-empty order book queues, i.e. $X_{t_i} \neq 0$,

(2) Order book at time t_i also contains the price-level for the mid-price $p_{t_{i+1}}$ at time t_{i+1}, i.e. $|p_{t_{i+1}} - p_{t_i}| \leqslant d \cdot \delta$, where δ is the tick-size.

In order to guarantee these two points, we have to consider the spread and the price-changes between consecutive order book states.

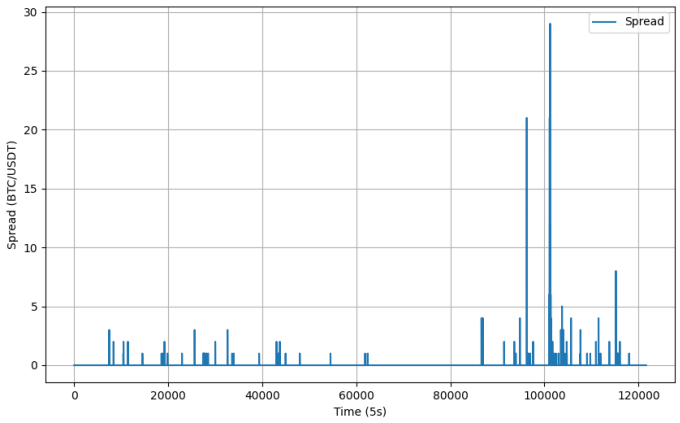


Figure 6: BTC/USDT Spread from 10.10.2023 to 17.10.2023

As indicated by Figure 6, the spread is 0 at most times, in fact approximately 99.9% of the time the spread is 0, which guarantees that the order book is not empty nearly always for any choice of d.

Figures 7 and 8 show shows the mid-price itself and the difference between consecutive order books states. As Figure 8 indicates, there were some flash crashes, causing the price of

[18]Many thanks to Tuan Thanh Truong for providing the data

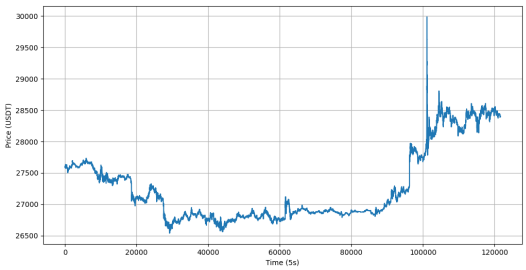


Figure 7: Mid-price evolution

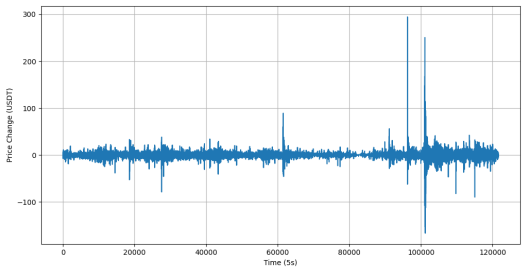


Figure 8: Mid-price difference

BTC/USDT to jump more than 100 USDT inside of a 5 second interval. However, it would not be a good idea to select d as 100 for this reason, as this representation increases memory and model complexity, whilst many levels would be empty. Instead, the empirical standard deviation σ of the mid-price difference is approximately 4 and we consider $2.5 \cdot \sigma = 10$ levels and are able to capture nearly 97% of the price changes.

Thus our order book representation is $O(t) = (X(t), p(t))$ with queue size $X_t = (-X^-, X^+) \in \mathbb{R}^{10} \times \mathbb{R}^{10}$ and mid-price $p_t \in \mathbb{R}$. It is also a common practice to normalize the data such that the features are on a comparable scale, in order to prevent overemphasis on one feature. This is also important for our dataset, as the mid-price is on a different scale than the queue sizes. For the mid-price we will use the Z-score normalization, which is defined for the empirical mean $\mu = \frac{1}{N}\sum_{j=1}^{N} p_{t_j}$ and standard-deviation $\sigma = \sqrt{\frac{1}{N}\sum_{j=1}^{N}(p_{t_j} - \mu)^2}$ as $\tilde{p}_{t_i} = \frac{p_{t_i} - \mu}{\sigma}$.

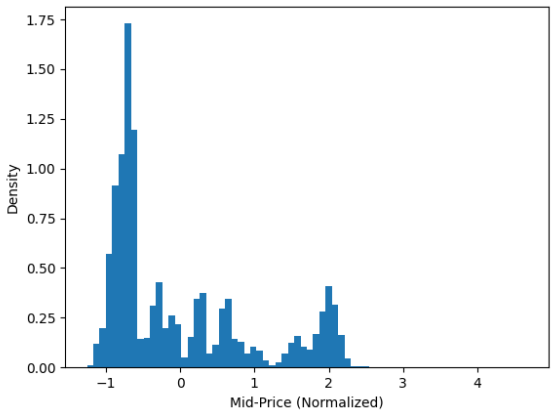


Figure 9: Density of normalized mid-price

On the other hand, the normalization of the queue sizes is a bit more difficult, as queue sizes around the levels further away from the mid-price tend to be zero. In the Wasserstein GAN model [12], Cont et al. suggest a square-root normalization, which we will also apply in a modified version. To preserve the sign of the queue, we choose the normalization as

$$\tilde{Q}_p(t) = \text{sign}(Q_p(t))\sqrt{|Q_p(t)|} + \text{sign}(Q_p(t)),$$

whereby we choose the last term, in order to create separation of the ask queue from the bid queue, so that the model also learns the difference and interplay between the ask and bid-side.

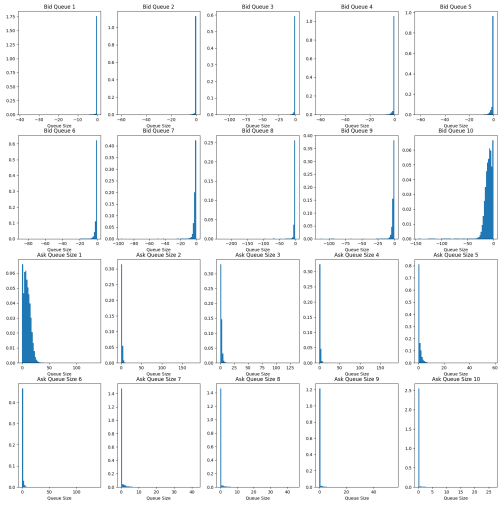


Figure 10: Density of queue size

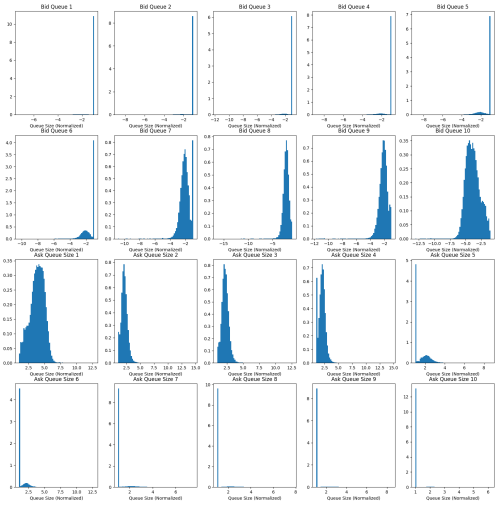


Figure 11: Density of normalized queue size

As demonstrated in Figures 10 and 11, the square-root normalization was able to transform the distribution so that there is less concentration at 0. However, starting from level 5 on the bid and queue size, the figures indicate that they are predominately empty, which is why we will only focus on the modelling the $d = 5$ closest levels to the mid-price, instead of 10.

3.4.2 Training and Validation

The training period consisted of six days of data from 10/10/2023 to 15/10/2023 and the following two days from 16/10/2023 to 17/10/2023 were used for validation. The model was trained for 30 epochs with each epoch consisting of 100 batches of 64 randomly sampled consecutive order book states. The former order book state was then used as an input to the diffusion model to predict the latter state.

For hyperparameter-tuning, the model was evaluated on the validation set. However, comparing the output of the diffusion model with the target variable may be inconclusive, since the output is probabilistic, which is why we use the so-called continuous ranked probability score that allows a comparison between a distribution and an observation.

Definition 3.14 (Continuous Ranked Probability Score (CRPS))**.** *Given a cumulative distribution function* $F\colon \mathbb{R} \to [0,1]$ *and an observation* $x \in \mathbb{R}$*, the continuous ranked probability score is defined as*

$$\mathrm{CRPS}(F, x) = \int_{\mathbb{R}} (F(z) - \mathbb{1}_{x \leqslant z})^2 \mathrm{d}z.$$

Note that we can generate many outputs using the diffusion model to estimate its empirical distribution. And since the output is 11-dimensional, the empirical distribution is estimated component-wise. Then the weighted sum of the individual scores is computed with higher weights assigned to the levels closer to the mid-price higher.

We then performed a grid search using the weighted CRPS on the validation set, which is demonstrated in the Appendix. The L2-loss of the objective defined in (15) using the selected hyper-parameters is shown below.

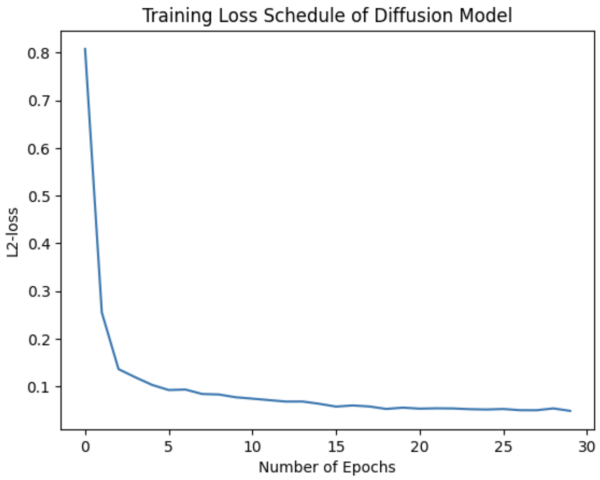


Figure 12: Training Loss schedule

The diffusion model is able to quickly learn the denoising process, but increasing the number of epochs did not improve the final loss as well as the validation score.

We are now ready to evaluate the trained model and do this on unseen order book data in the two-day period from 18/10/2023 to 19/10/2023.

4 Evaluation

We begin by comparing marginal distributions of generated and actual queue sizes as well as mid-prices. The marginal distribution is estimated by randomly sampling 1000 states O_t and generating the following state $O_{t+\Delta t}$. As seen in Figure 13, the marginal distributions of the queue sizes are learnt quite well. During the experiments, there were a few outliers, which could be an issue, when generating sample-paths in an auto-regressive manner. Therefore, a clip function was applied that limits the range of the output within $[-10, 10]$.

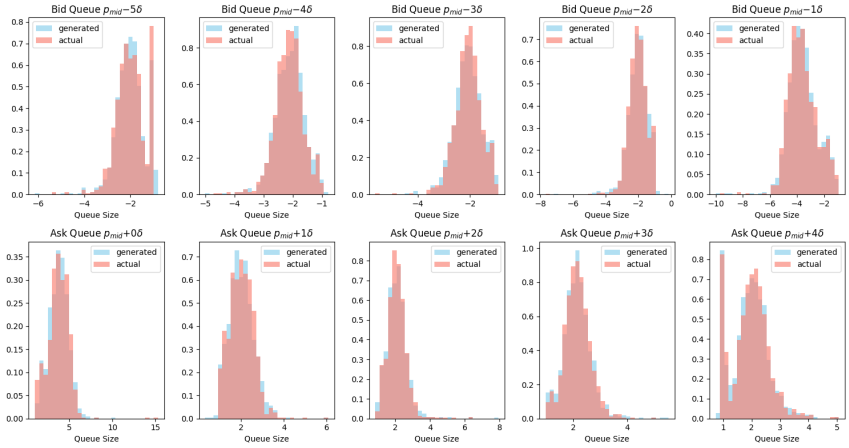


Figure 13: Marginal distribution of queue sizes

However, the marginal distribution itself is not a good indicator, as simply predicting the order book state $O_{t+\Delta t}$ as O_t would also yield positive results. To verify that the diffusion model has variation in its output, we randomly sample a window of the time-series and at each time generate 100 samples of the next state. Then the quantiles of the generated sample states were compared to the observed state. Ideally, the observed state is contained within the quantiles of the generated samples. This was indeed the case for various sampled windows, where Figure 14 shows an example for the queue size.

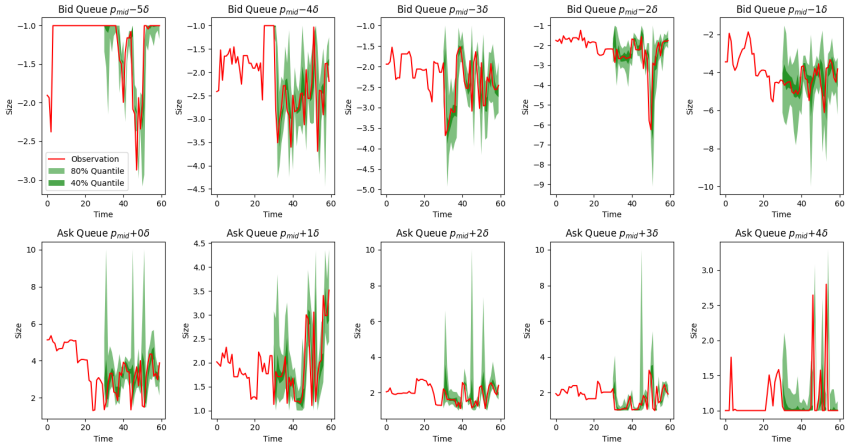


Figure 14: 40th and 80th percentile of generated queue size

The same observation holds true for the mid-price, which was largely within the 80th quantiles of various sampled windows, as seen by example in Figure 15.

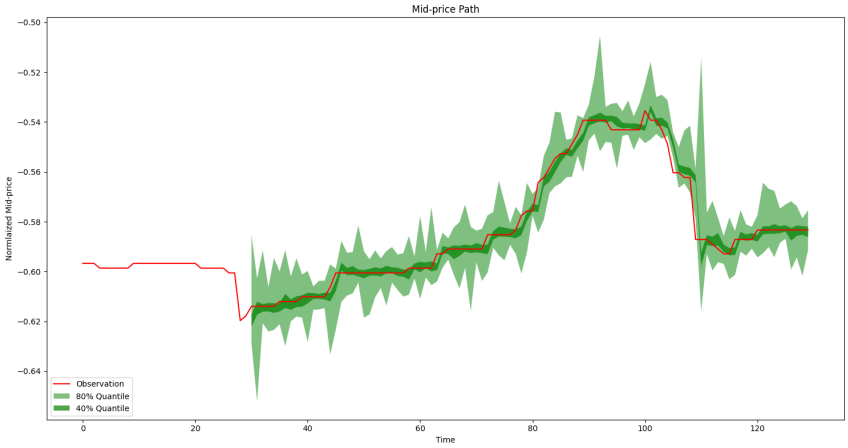


Figure 15: 40th and 80th percentile of generated mid-price

The previous insights suggest that generated order book states one time period $\Delta t = 5$ sec. in the future align well with the observed states. We will now evaluate an entire price path that is generated by the diffusion model, which can be achieved auto-regressively, i.e. using a generated state as input for the model to generate the next state. Figure 16 shows 20 price paths of actual and auto-regressively generated price paths over 100 time steps. The initial state was randomly sampled from the test set. Note that the starting value of each path was subtracted from each path, so that these start in 0.

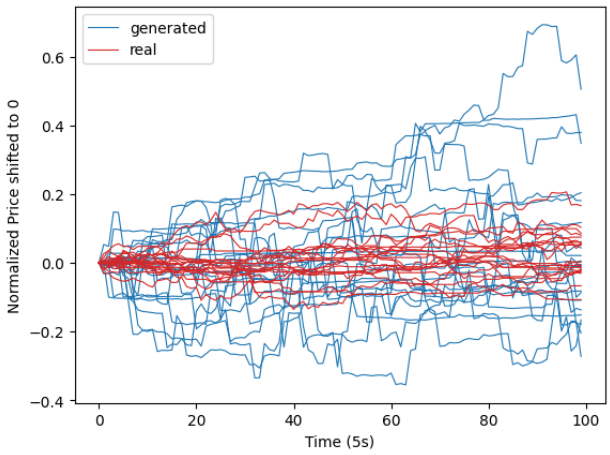


Figure 16: Actual vs generated price paths

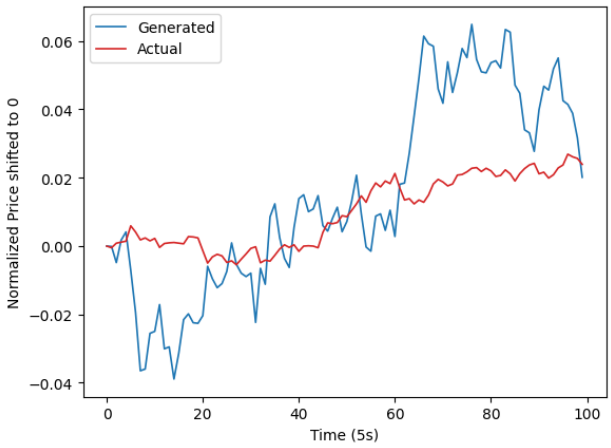


Figure 17: Mean price paths

The generated price path seems to be more volatile than the observed price paths. Instead of sampling only one state at each time, we therefore sample three states and select the median of these. Figure 18 and 19 show the resulting paths.

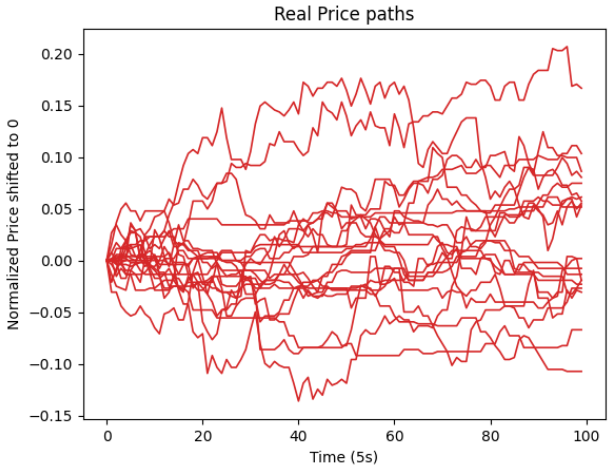


Figure 18: Actual price paths

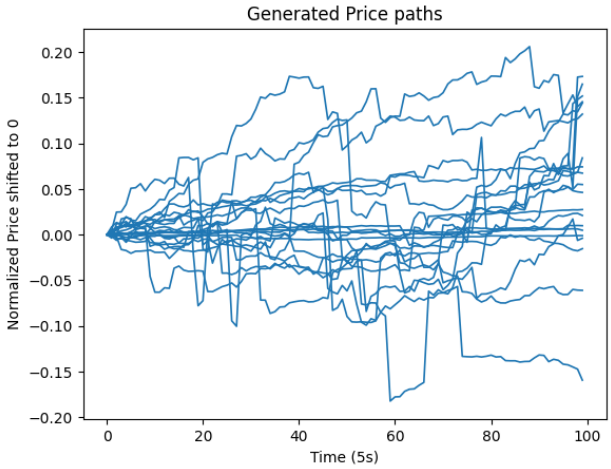


Figure 19: Generated price paths (3 samples)

Using the approach of sampling three states and selecting the median reduced the overall volatility of the paths. Figure 20 and 21 further support that these generated paths are more similar to the observed paths.

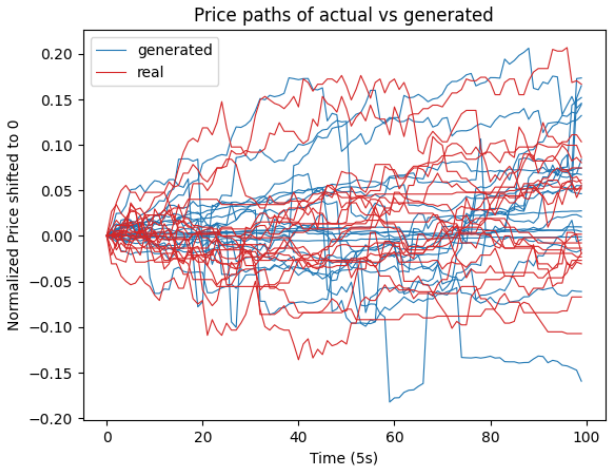


Figure 20: Actual vs generated price paths

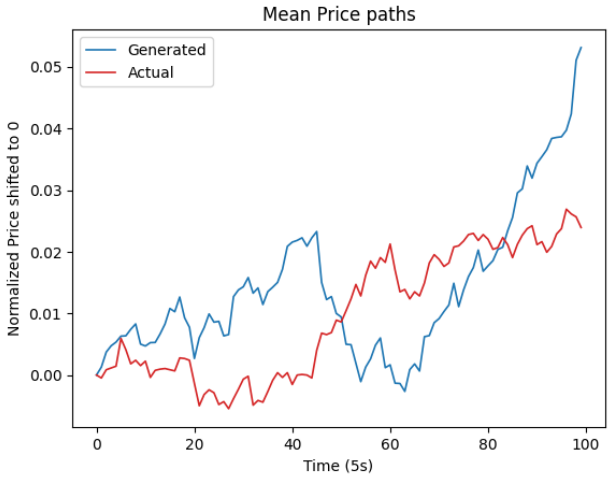


Figure 21: Mean price paths

Finally, we generate six hours of a price path (4320 data-points) and evaluate these based on known empirical facts about price paths also known as stylized facts [42]. Figure 22 shows the generated path, where only one sample was generated per time-step. Figure 23 uses the median of three samples. As expected, the path using a single sample is more volatile and therefore we only evaluate the latter path using the stylized facts.

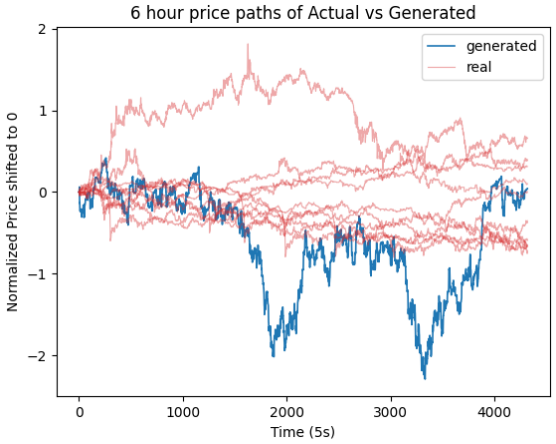


Figure 22: Generated path (1 sample)

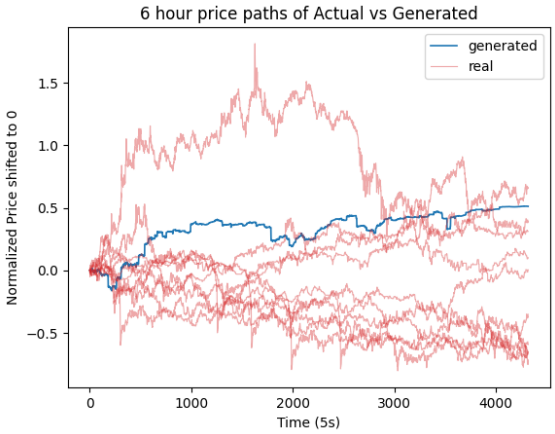


Figure 23: Generated path (Median of 3)

Stylized facts

(1) Distribution of log-returns: For a given time-scale $\Delta t > 0$ and mid price process $(p_{\text{mid}}(t))_{t\in[0,T]}$, the log-return at time scale Δt is defined as

$$r(t, \Delta t) = \log\left(\frac{p_{\text{mid}}(t+\Delta t)}{p_{\text{mid}}(t)}\right).$$

In the Black-Scholes model, log-returns are normally distributed. Empirical data also indicates that log-returns exhibit normality over longer time-scales Δt, while smaller Δt often display heavy tails, which can be measured by kurtosis. For a random variable $X \in \mathcal{L}^4(\mathbb{P})$ with mean $\mu \in \mathbb{R}$ and standard deviation $\sigma > 0$, its kurtosis is defined as

$$K(X) := \mathbb{E}\left[\left(\frac{X-\mu}{\sigma}\right)^4\right] - 3.$$

If X is normally distributed, its kurtosis is 0. Figures 24 and 25 show the log-return distribution for different time scale $\Delta t \in \{5 \text{ sec.}, 10 \text{ sec.}, 1 \text{ min.}, 5 \text{ min.}\}$. For small Δt, the true return distribution is dominated by 0 values, whilst the generated return distribution seems closer to a normal distribution, which is also confirmed in Table 2.

Time-scale (Δt)	**True Return Kurtosis**	**Generated Return Kurtosis**
`5s`	483	191
`30s`	470	31
`1min`	632	14
`5min`	614	2
`30min`	56	N/A
`1h`	14	N/A

Table 2: Comparison of Kurtosis for true and generated returns

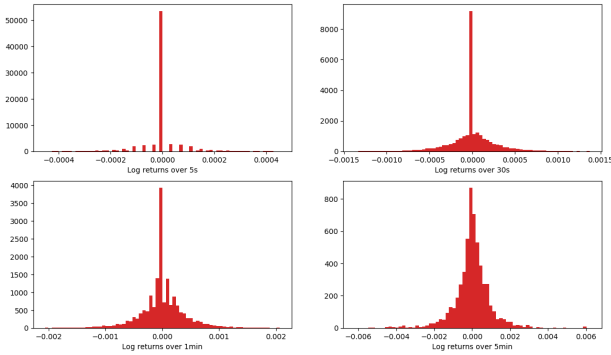


Figure 24: Actual log-return distributions

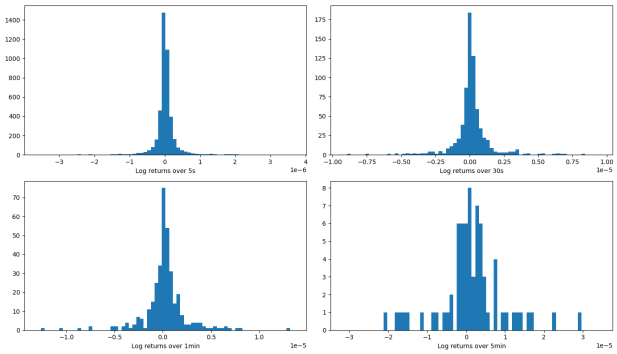


Figure 25: Generated log-return distributions

The generated returns seem to exhibit normality at small-time scales as well, yet the kurtosis in Table 2 indicates that these also have a heavier-tailed distribution than the normal distribution, which is also the case for the true distribution. The decrease in kurtosis over the time scale Δt is present in both the actual and generated returns however, the generated returns seem to tend to a normal behavior at a faster rate.

(2) Absence of autocorrelation: The autocorrelation function C is defined for some time lag τ and log-return process $r(t, \Delta t)$ with time-scale Δt as

$$C(\tau) = \text{corr}(r(t, \Delta t), r(t + \tau, \Delta t)).$$

In other words, it is the correlation between the log-returns at time-scale Δt and itself shifted by τ. Empirically, the autocorrelation rapidly decays to 0 and one explanation of this is that otherwise there exists strategies that can take advantage of this positive correlation, which is known as statistical arbitrage. This would in turn lead to market corrections and reduce the autocorrelation [9], which also supports the so-called efficient market hypothesis that states that all available information is already priced into a security. The decay of autocorrelation of log-returns with different time shifts τ was also present for the BTC/USDT data, as seen in Figure 26. The generated returns were also able to replicate this phenomenon as depicted in Figure 27.

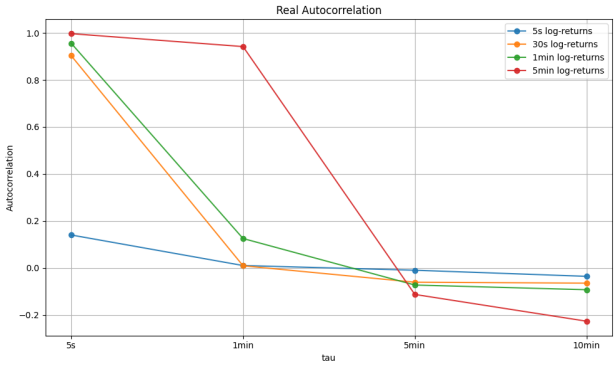


Figure 26: Real Autocorrelation

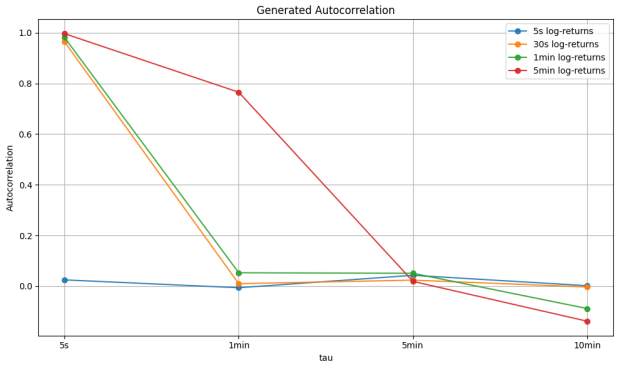


Figure 27: Generated Autocorrelation

(3) Volatility Clustering: Random-walk models and in general the returns of any price process with independent increments do not exhibit autocorrelation. Moreover any (measurable) function of the increments also remains independent and therefore does not have autocorrelation. Yet stock returns typically exhibit some autocorrelation between absolute returns. This phenomena is also known as volatility clustering and implies that periods of high and low volatility typically persist for period of time. An extreme example of this was the COVID-19 pandemic, where there was high market uncertainty that caused the S&P 500 index to be highly volatile over a long period.

This phenomenon can be measured by applying the autocorrelation function to the absolute returns

$$C(\tau) = \text{corr}(|r(t, \Delta t)|, |r(t + \tau, \Delta t)|).$$

The actual returns confirm this stylized property, as the correlation of the absolute returns persisted even for longer time shifts τ. However, the generated returns unfortunately did not exhibit this behavior as seen in Figure 28 and 29.

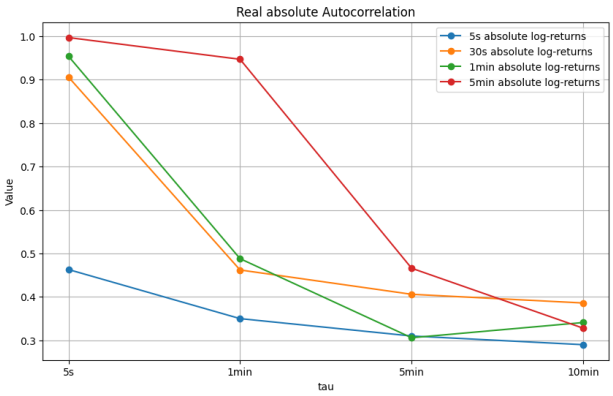


Figure 28: Real abs Autocorrelation

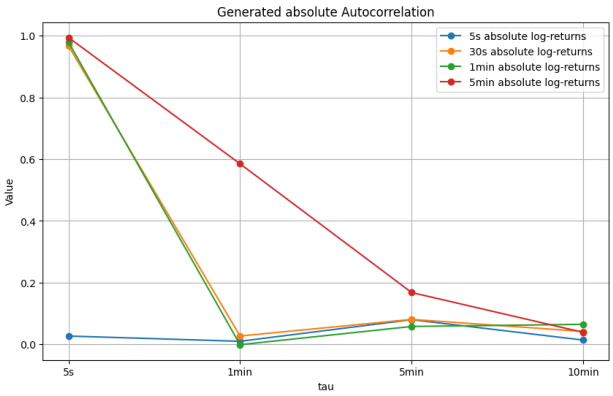


Figure 29: Generated abs Autocorrelation

5 Conclusion

During the last months of working on this thesis, I learned a lot about limit order book modelling and acquired additional programming skills. Without any prior understanding of diffusion models, I can confidently say that the training and sampling process of diffusion models do not seem like black-box algorithms to me anymore (and hopefully to the reader). Furthermore, the literature review on existing models was very fascinating for me to see the creative approaches, specifically the framework I reviewed in Section 1. Upon reflection, some tasks such as the training of the model could have been done more time-efficient by for example using a GPU instead. Additionally, I thought that the proof of Theorem 3.13 with the ergodicity of Kernels would be completely analogous for a sequence of transition Kernels, but only after proof-reading and finally writing down the proof, I realized that the additional assumption of the existence of $N \in \mathbb{N}$ such that $K_n = K_m$ for $n \geqslant N$ was necessary for this proof. The existence of the convergent subsequence in the proof also requires further considerations. This was an important lesson to be particularly cautious, when using generalized results and checking for minor details. Overall this has been a valuable learning experience. To summarize and reflect on the results of this thesis.

The proposed diffusion model was able to capture the marginal distributions of the queue sizes and mid-prices (Figure 13) and generate samples that are visually consistent with observations (Figures 14, 15). The generated price paths replicated the stylized facts to a certain degree. The log-returns at short time-scales Δt had heavier tails than the normal distribution measured by kurtosis (Table 2), which was consistent with the actual returns. Furthermore, the generated log-returns did not have any autocorrelation (Figure 27), but was unable to replicate the volatility clustering behavior (Figure 29). This may be because the TimeGrad architecture only encoded the previous order book state in the hidden state (19). Therefore, it may be interesting to include a longer history as an additional variable. This is furthermore supported by Sirigano and Cont [36], who found that order books exhibit long-range dependence and including a longer history can improve deep learning models.

Additionally, further methods of data-processing perhaps should have been considered. BTC/USDT is highly volatile at periods, which was present in the training data and may explain why the generated price process with only one sample generated paths that were more volatile than the actual price processes (Figure 22). Such periods of extreme volatility could therefore be removed from the training set. Instead of modelling the mid-prices with a seemingly difficult distribution to learn (Figure 11), one could also consider modelling returns, as the mid-prices can be derived from the returns given an initial price. Log-returns over larger time-scales also exhibit nicer distributions (Figure 24). From a computational point of view, diffusion models are also not feasible without high-speed GPUs, as the inference on a Apple M1 CPU takes around two seconds per generation. This could be overcome generating an entire prediction window at once, instead of an auto-regressive approach. For example, Shen and Kwok (2023) [34] recently proposed a diffusion model known as TimeDiff, which samples an entire prediction interval immediately. This could all be considered for future work, as the desire for order book models will certainly advance this research area forward in the coming years, and I am also interested in continuing to work on this project.

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Appendix

Proofs in Section 3

For the constructed forward process (3) with natural filtration $(\mathcal{F}_t)_{t=0,\dots,T}$ the following equality holds true

$$\mathbb{E}\left[f(\sqrt{1-\beta_{t+1}}Z_t+\sqrt{\beta_{t+1}}\varepsilon_t)\mid\sigma(Z_t)\right]=\mathbb{E}\left[f(\sqrt{1-\beta_{t+1}}Z_t+\sqrt{\beta_{t+1}}\varepsilon_t)\mid\mathcal{F}_t\right].$$

Proof. Similar to Corollary 3.6, we use the alternative characterization of the conditional expectation. Let Z be $\mathcal{F}_t=\sigma(Z_i\colon i=0,\dots,t)$-measurable and $g\colon(\mathbb{R}^d)^{\otimes t}\to\mathbb{R}^d$ such that $Z=g(Z_0,\dots,Z_t)$ holds. Then, we have

$$\begin{aligned}
&\mathbb{E}\left[g(Z_0,\dots,Z_t)f(\sqrt{1-\beta_{t+1}}Z_t+\sqrt{\beta_{t+1}}\varepsilon_t)\right]\\
&=\int_{\mathbb{R}^d}g(z_0,\dots,z_t)f(\sqrt{1-\beta_{t+1}}z_t+\sqrt{\beta_{t+1}}y_t))\mathbb{P}_{(Z_0,\dots,Z_t,\varepsilon_t)}(\mathrm{d}(z_0,\dots,z_t,y_t))\\
&=\int_{\mathbb{R}^d}\int_{\mathbb{R}^d}g(z_0,\dots,z_t)f(\sqrt{1-\beta_{t+1}}z_t+\sqrt{\beta_{t+1}}y_t))\mathbb{P}_{(Z_0,\dots,Z_t)}(\mathrm{d}(z_0,\dots,z_t))\otimes\mathbb{P}_{\varepsilon_t}(\mathrm{d}y_t)\\
&=\int_{\mathbb{R}^d}g(z_0,\dots,z_t)\int_{\mathbb{R}^d}f(\sqrt{1-\beta_{t+1}}z_t+\sqrt{\beta_{t+1}}y_t))\mathbb{P}_{\varepsilon_t}(\mathrm{d}y_t)\otimes\mathbb{P}_{(Z_0,\dots,Z_t)}(\mathrm{d}(z_0,\dots,z_t))\\
&=\mathbb{E}\left[g(Z_0,\dots,Z_t)\mathbb{E}_{\varepsilon_t}[f(\sqrt{1-\beta_{t+1}}Z_t+\sqrt{\beta_{t+1}}\varepsilon_t)]\right],
\end{aligned}$$

where the independence of $(Z_0,\dots,Z_t)$ and ε_t was used in the third equality and Fubini/Tonelli in the fourth. Thus, $\mathbb{E}_{\varepsilon_t}[f(\sqrt{1-\beta_{t+1}}Z_t+\sqrt{\beta_{t+1}}\varepsilon_t)]=\mathbb{E}\left[f(\sqrt{1-\beta_{t+1}}Z_t+\sqrt{\beta_{t+1}}\varepsilon_t)\mid\mathcal{F}_t\right]$, which proves the desired equality, as Corollary 3.6 shows that $\mathbb{E}_{\varepsilon_t}[f(\sqrt{1-\beta_{t+1}}Z_t+\sqrt{\beta_{t+1}}\varepsilon_t)]=\mathbb{E}\left[f(\sqrt{1-\beta_{t+1}}Z_t+\sqrt{\beta_{t+1}}\varepsilon_t)\mid\sigma(Z_t)\right]$. □

Theorem 3.13. *A sequence of strictly positive and regular transition kernels $(K_n)_{n\in\mathbb{N}}$ with $K_n=K_m$ for $n,m\geqslant N$ and $N\in\mathbb{N}$ are ergodic if there exists a strictly positive stationary distribution α.*

Proof. Let α be a strictly positive stationary distribution and we denote with $\mathbb{E}_\alpha[\cdot]$, the expectation w.r.t. α. Furthermore, let u_0 be a bounded and continuous function on $\mathbb{R}^d$ with transformations $(u_n)_{n\in\mathbb{N}_{>0}}$ defined as

$$u_n(x):=\int_{\mathbb{R}^d}u_{n-1}(y)K_n(\mathrm{d}y,x). \tag{20}$$

Then for $n\in\mathbb{N}_{>0}$, we have

$$\begin{aligned}
\mathbb{E}_\alpha[u_n]&=\int_{\mathbb{R}^d}\int_{\mathbb{R}^d}u_{n-1}(y)K_n(\mathrm{d}y,x)\alpha(\mathrm{d}x)\\
&=\int_{\mathbb{R}^d}u_{n-1}(y)\int_{\mathbb{R}^d}K_n(\mathrm{d}y,x)\alpha(\mathrm{d}x)\\
&=\int_{\mathbb{R}^d}u_{n-1}(y)\alpha(\mathrm{d}y)=\mathbb{E}_\alpha[u_{n-1}],
\end{aligned}$$

where we use the fact that α is a stationary distribution in the third equality. Additionally, $\mathbb{E}_\alpha[|u_n|]$ decreases in n, which is easily seen as follows

$$\mathbb{E}_\alpha[|u_{n+1}|] = \int_{\mathbb{R}^d} \left| \int_{\mathbb{R}^d} u_n(y) K_{n+1}(\mathrm{d}y, x) \right| \alpha(\mathrm{d}x) \leqslant \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |u_n(y)| K_{n+1}(\mathrm{d}y, x) \alpha(\mathrm{d}x) = \mathbb{E}_\alpha[|u_n|]. \tag{21}$$

Thus $\lim_{n\to\infty} \mathbb{E}_\alpha[|u_n|] := m$ exists. If u_0 is absolutely continuous, then by the regularity of the kernels $(u_n)_{n\in\mathbb{N}}$ is equicontinuous. And by Arzela-Ascoli or (Thm.3, Feller pp.270, [16]), there exists a convergent sub-sequence such that $(u_{n_k})_{k\in\mathbb{N}}$ converges point-wise against a function v_0, i.e. $u_{n_k} \to v_0$ for $k \to \infty$. However, Feller (pp.273 [16]) asserts that there also exists a convergent sub-sequence for continuous and bounded functions, which we assume in the following (will be discussed at the end of the proof). Therefore we also have $u_{n_k+1} \to v_1$, where v_1 is the transform of v_0 defined in (20) with kernel K_N. This is why we need the assumption that there exists $N \in \mathbb{N}$ such that $K_n = K_m$ for $n, m \geqslant N$.

It is easily verified that $\max_{x\in\mathbb{R}^d} |u_n(x)| \leqslant \max_{x\in\mathbb{R}^d} |u_m(x)|$ holds true for $n > m$. By dominated convergence, we have $\mathbb{E}_\alpha[u_n] \to \mathbb{E}_\alpha[v_0]$ and $\mathbb{E}_\alpha[|u_n|] \to \mathbb{E}_\alpha[|v_0|]$. As a result of that

$$\mathbb{E}_\alpha[u_0] = \mathbb{E}_\alpha[v_0] = \mathbb{E}_\alpha[v_1] \quad \text{and} \quad \mathbb{E}_\alpha[|v_0|] = \mathbb{E}_\alpha[|v_1|] \tag{22}$$

hold true. As the kernels are strictly positive and v_0 is continuous, the last equality implies that v_0 cannot change signs, as otherwise there would be a strict inequality in (21). If $\mathbb{E}_\alpha[u_0] = 0$ holds, the first equality in (22) implies that $v_0 \equiv 0$. Consequently for any $u_0 \in \mathcal{C}_b(\mathbb{R}^d)$, we have $v_0 \equiv \mathbb{E}_\alpha[u_0]$ by considering the function $u_0 - \mathbb{E}_\alpha[u_0]$. This also implies that any transformations of v_0 is also constant and it follows that $u_n(x) \to v_0(x) = \mathbb{E}_\alpha[u_0]$ for $n \to \infty$ and any $x \in \mathbb{R}^d$. Writing the transition kernel after n steps as

$$K^n(A, x) = \int_{\mathbb{R}^d} \dots \int_{\mathbb{R}^d} \mathbb{1}_A(x_0) K_1(\mathrm{d}x_0, x_1) \dots K_n(\mathrm{d}x_{n-1}, x),$$

allows us to write $u_n(x) = \int_{\mathbb{R}^d} u_0(y) K^n(\mathrm{d}y, x)$. Additionally $u_n(x) \to \mathbb{E}_\alpha[u_0]$ holds for any bounded and continuous function u_0, which implies the convergence in distribution and by Portmanteau's lemma[19] this is equivalent to $K^n(A, x) \to \alpha(A)$ for continuity sets, i.e. sets where the boundary ∂A is a null-set $\alpha(\partial A) = 0$. For any initial distribution γ_0, the distribution of the forward process after n steps is given by

$$\gamma_n(A) = \int_{\mathbb{R}^d} K^n(A, x) \gamma_0(dx),$$

which is verified by recursively substituting γ_n in (7). Letting n tend to infinity and by dominated convergence, we have

$$\lim_{n\to\infty} \gamma_n(A) = \lim_{n\to\infty} \int_{\mathbb{R}^d} K^n(A, x) \gamma_0(dx) = \int_{\mathbb{R}^d} \lim_{n\to\infty} K^n(A, x) \gamma_0(dx) = \alpha(A).$$

[19] Prof. König lecture notes, pp.124 https://www.wias-berlin.de/people/koenig/www/WTSkript.pdf

for all continuity sets $A \in \mathcal{B}(\mathbb{R}^d)$. By Portmanteau's lemma, we conclude the convergence in distribution, i.e. $\gamma_n \xrightarrow{d} \alpha_0$ for $n \to \infty$

Note: The assumption of the existence of a converging sub-sequence of u_0 was important to justify the convergence in distribution of $K^n(A, x) \to \alpha(A)$. Feller references the previous Theorem, where it was additionally assumed that Ω was a bounded and closed interval, which implies that any continuous function u_0 on Ω is already uniformly continuous. Nevertheless, this requires further investigation and perhaps additional regularity constraints. □

The proof of Equation (16) can be found in a similar form in [40]

$$\min_{\theta\in\Theta} \mathbb{E}_{q(x)} \left[\mathbf{d}_{\mathrm{KL}} \left(q(z \mid x) \| p_\theta(z \mid x)\right)\right] = \min_{\theta\in\Theta} -\mathbb{E}_{q(x)p_\theta(z|x)} \left[\log \frac{p_\theta(x, z)}{q(z \mid x)}\right].$$

Proof. We have

$$\begin{aligned}
\log q(x) &= \mathbb{E}_{p_\theta(z|x)}[\log q(x)] \\
&= \mathbb{E}_{p_\theta(z|x)} \left[\log \frac{q(x, z)}{q(z \mid x)}\right] \\
&= \mathbb{E}_{p_\theta(z|x)} \left[\log \frac{q(x, z) p_\theta(z \mid x)}{q(z \mid x) p_\theta(z \mid x)}\right] \\
&= \mathbb{E}_{p_\theta(z|x)} \left[\log \frac{q(x, z)}{p_\theta(z \mid x)}\right] + \mathbb{E}_{p_\theta(z|x)} \left[\log \frac{p_\theta(z \mid x)}{q(z \mid x)}\right] \\
&= -\mathbb{E}_{p_\theta(z|x)} \left[\log \frac{p_\theta(z \mid x)}{q(x, z)}\right] - \mathbf{d}_{\mathrm{KL}} \left(q(z \mid x) \| p_\theta(z \mid x)\right).
\end{aligned}$$

Rearranging and taking the expectation w.r.t. $q(x)$ completes the proof

$$\mathbb{E}_{q(x)} \left[\log q(x) + \mathbf{d}_{\mathrm{KL}} \left(q(z \mid x) \| p_\theta(z \mid x)\right)\right] = -\mathbb{E}_{q(x)p_\theta(z|x)} \left[\log \frac{p_\theta(z \mid x)}{q(x, z)}\right],$$

as the $\log q(x)$ term does not depend on θ. □

Proposition 3.16. *For the upper bound from Proposition 3.15, the following equality holds*

$$\begin{aligned}
-\mathbb{E}_{q(x_{0:T})} \left[\log \frac{p_\theta(x_0, x_{1:T})}{q(x_{1:T} \mid x_0)}\right] &= \mathbb{E}_{q(x_{0:T})} \left[-\log p_\theta\left(x_0 | x_1\right)\right] \\
&\quad + \sum_{t=2}^{T} \mathbb{E}_{q(x_0, x_t)} \left[\mathbf{d}_{\mathrm{KL}} \left(q(x_{t-1} \mid x_0, x_t) \| p_\theta(x_{t-1} | x_t)\right)\right] \\
&\quad + \mathbb{E}_{q(x_0, x_T)} \left[\mathbf{d}_{\mathrm{KL}} \left(q(x_T | x_0) \| p_\theta(x_T)\right)\right]
\end{aligned}$$

Proof. By the Markov property on the conditional densities, we have

$$\begin{aligned}
\mathbb{E}_{q(x_{0:T})}\left[\log \frac{p_\theta(x_0, x_{1:T})}{q(x_{1:T} \mid x_0)}\right] &= \mathbb{E}_{q(x_0, x_{1:T})}\left[\log \frac{p_\theta(x_T)\prod_{t=1}^{T} p_\theta\left(x_{t-1}|x_t\right)}{\prod_{t=1}^{T} q\left(x_t|x_0, x_{t-1}\right)}\right] \\
&= \mathbb{E}_{q(x_{0:T})}\left[\log p_\theta(x_T) + \sum_{t=1}^{T} \log \frac{p_\theta\left(x_{t-1}|x_t\right)}{q\left(x_t|x_0, x_{t-1}\right)}\right] \\
&= \mathbb{E}_{q(x_{0:T})}\left[\log p_\theta(x_T) + \log \frac{p_\theta\left(x_0|x_1\right)}{q\left(x_1|x_0\right)} + \sum_{t=2}^{T} \log \frac{p_\theta\left(x_{t-1}|x_t\right)}{q\left(x_t|x_0, x_{t-1}\right)}\right].
\end{aligned} \tag{23}$$

Now consider the following identity

$$\sum_{t=2}^{T} \log \frac{q(x_t \mid x_0, x_{t-1})}{q(x_{t-1} \mid x_0, x_t)} = \sum_{t=2}^{T} \log \frac{q(x_0, x_t)}{q(x_0, x_{t-1})} = \log q(x_0, x_T) - \log q(x_0, x_1) = \log \frac{q(x_T \mid x_0)}{q(x_1 \mid x_0)},$$

since the second sum is a telescoping sum. So adding an additive 0 to equation (18) yields

$$\begin{aligned}
&\mathbb{E}_{q(x_{0:T})}\left[\log p_\theta(x_T) + \log \frac{p_\theta\left(x_0|x_1\right)}{q\left(x_1|x_0\right)} + \sum_{t=2}^{T} \log \frac{p_\theta\left(x_{t-1}|x_t\right)}{q\left(x_t|x_0, x_{t-1}\right)}\right] \\
&+ \mathbb{E}_{q(x_{0:T})}\left[\sum_{t=2}^{T} \log \frac{q(x_t \mid x_0, x_{t-1})}{q(x_{t-1} \mid x_0, x_t)} - \log \frac{q(x_T \mid x_0)}{q(x_1 \mid x_0)}\right] \\
&= \mathbb{E}_{q(x_{0:T})}\left[\log p_\theta\left(x_0|x_1\right) + \sum_{t=2}^{T}\left(\log \frac{p_\theta\left(x_{t-1}|x_t\right)}{q\left(x_t|x_0, x_{t-1}\right)} + \log \frac{q(x_t \mid x_0, x_{t-1})}{q(x_{t-1} \mid x_0, x_t)}\right) + \log \frac{p_\theta(x_T)}{q(x_T \mid x_0)}\right] \\
&= \mathbb{E}_{q(x_{0:T})}\left[\log p_\theta\left(x_0|x_1\right) + \sum_{t=2}^{T} \log \frac{p_\theta\left(x_{t-1}|x_t\right)}{q(x_{t-1} \mid x_0, x_t)} + \log \frac{p_\theta(x_T)}{q(x_T \mid x_0)}\right] \\
&= \mathbb{E}_{q(x_{0:T})}\left[\log p_\theta\left(x_0|x_1\right)\right] - \sum_{t=2}^{T} \mathbb{E}_{q(x_0, x_t)}\left[\mathbf{d}_{\mathrm{KL}}\left(q(x_{t-1} \mid x_0, x_t) \| p_\theta(x_{t-1}|x_t)\right)\right] \\
&- \mathbb{E}_{q(x_0, x_T)}\left[\mathbf{d}_{\mathrm{KL}}\left(q(x_T|x_0) \| p_\theta(x_T)\right)\right].
\end{aligned}$$

Taking the negative of both sides yields the desired equality. □

Proposition 3.19. *Assuming that the model distribution $p_\theta(x_{t-1} \mid x_t)$ is Gaussian, then for the optimization objective L defined in (11), the following equality holds true*

$$L(\theta) = \sum_{t=2}^{T} \frac{1}{2\sigma_q^2(t)} \frac{\left(1-\alpha_t\right)^2}{\alpha_t\left(1-\bar{\alpha}_t\right)} \cdot \mathbb{E}\left[\left\|\varepsilon_t - \varepsilon_\theta\left(x_t, t\right)\right\|_2^2\right].$$

Proof. The proof of this Proposition can also be found on page 15 of [26]. However as the proof is rather long and in order to remain consistent with the notation, we will include this.

We will use the fact that the KL-divergence between two multivariate Gaussian distributions is given by

$$\mathbf{d}_{\mathrm{KL}}\left(\mathcal{N}\left(x;\mu_x,\Sigma_x\right)\|\mathcal{N}\left(y;\mu_y,\Sigma_y\right)\right)=\frac{1}{2}\left(\log\frac{|\Sigma_y|}{|\Sigma_x|}-d+\operatorname{tr}\left(\Sigma_y^{-1}\Sigma_x\right)+\left(\mu_y-\mu_x\right)^T\Sigma_y^{-1}\left(\mu_y-\mu_x\right)\right).$$

Now the proof is straightforward, but requires calculation. As a reminder, we have for $t=2,\ldots,T$

$$q(x_{t-1}\mid x_t,x_0)=f_{\mathcal{N}}\left(x_{t-1};\underbrace{\frac{1}{\sqrt{\alpha_t}}\left(x_t-\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_t\right)}_{:=\mu_q(t)},\underbrace{\frac{\left(1-\alpha_t\right)\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}}_{:=\sigma_q^2(t)}\mathrm{I}_d\right)$$

with $\Sigma_q(t):=\sigma_q^2(t)\mathrm{I}_d$ and

$$p_\theta(x_{t-1}\mid x_t)=f_{\mathcal{N}}\left(x_{t-1};\underbrace{\frac{1}{\sqrt{\alpha_t}}\left(x_t-\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta(t)\right)}_{:=\mu_\theta(t)},\Sigma_q(t)\right).$$

Then for the objective L, we calculate

$$\begin{aligned}
L(\theta)&=\sum_{t=2}^{T}\mathbb{E}_{q(x_{0:T})}\left[\mathbf{d}_{\mathrm{KL}}\left(q(x_{t-1}\mid x_0,x_t)\|p_\theta(x_{t-1}|x_t)\right)\right]\\
&=\frac{1}{2}\sum_{t=2}^{T}\mathbb{E}_{q(x_{0:T})}\left[\log\frac{|\Sigma_q(t)|}{|\Sigma_q(t)|}-d+\operatorname{tr}\left(\Sigma_q(t)^{-1}\Sigma_q(t)\right)+\left(\mu_\theta(t)-\mu_q(t)\right)^T\Sigma_q(t)^{-1}\left(\mu_\theta(t)-\mu_q(t)\right)\right]\\
&=\frac{1}{2}\sum_{t=2}^{T}\mathbb{E}_{q(x_{0:T})}\left[\left(\mu_\theta(t)-\mu_q(t)\right)^T\Sigma_q(t)^{-1}\left(\mu_\theta(t)-\mu_q(t)\right)\right]\\
&=\frac{1}{2}\sum_{t=2}^{T}\mathbb{E}_{q(x_{0:T})}\left[\left(\mu_\theta(t)-\mu_q(t)\right)^T\left(\sigma_q^2(t)\mathrm{I}_d\right)^{-1}\left(\mu_{\boldsymbol{\theta}}(t)-\mu_q(t)\right)\right]\\
&=\sum_{t=2}^{T}\frac{1}{2\sigma_q^2(t)}\mathbb{E}_{q(x_{0:T})}\left[\|\mu_\theta(t)-\mu_q(t)\|_2^2\right]\\
&=\sum_{t=2}^{T}\frac{1}{2\sigma_q^2(t)}\mathbb{E}_{q(x_{0:T}),\varepsilon_t}\left[\left\|\frac{1}{\sqrt{\alpha_t}}\left(\left(x_t-\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_t\right)-\left(x_t-\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta(x_t,t)\right)\right)\right\|_2^2\right]\\
&=\sum_{t=2}^{T}\frac{1}{2\sigma_q^2(t)}\mathbb{E}_{q(x_{0:T}),\varepsilon_t}\left[\left\|\frac{1}{\sqrt{\alpha_t}}\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\left(\varepsilon_\theta(x_t,t)-\varepsilon_t\right)\right\|_2^2\right]\\
&=\sum_{t=2}^{T}\frac{1}{2\sigma_q^2(t)}\frac{(1-\alpha_t)^2}{\alpha_t(1-\bar{\alpha}_t)}\mathbb{E}\left[\|(\varepsilon_t-\varepsilon_\theta(X_t,t))\|_2^2\right].
\end{aligned}$$

□

Hyper-parameter tuning

Figure 30 shows the results of the grid search and hyper-parameter tuning. In the table `cell type` refers to the model to encode the hidden state seen in (16) with n number of layers (`num layers`). `diff steps` refer to the number of steps in the forward process and `beta end` the last value in the variance schedule. Lastly, `lr` is the learning rate and the invalid constructions refer to generated outputs of the diffusion model, where the bid-queue has values greater than -1 and the ask-queue has values less than 1, which should not happen by construction.

cell type	num layers	diff steps	beta end	lr	Training Loss	Invalid Constructions	Weighted CRPS
GRU	2	80	0.1	1e-2	0.0534	2090	0.2069
GRU	2	80	0.1	1e-3	0.0485	449	0.1610
GRU	2	80	0.1	1e-4	0.052	299	0.1729
GRU	2	80	0.1	1e-5	0.0487	327	0.1819
GRU	2	80	0.1	1e-6	0.0511	591	0.1642
GRU	2	80	0.01	1e-2	0.0523	708	0.2193
GRU	2	80	0.01	1e-3	/	413	0.1737
GRU	2	80	0.01	1e-4	/	337	0.1593
GRU	2	80	0.01	1e-5	/	627	0.1646
GRU	2	80	0.01	1e-6	0.0512	943	0.1785
GRU	2	100	0.1	1e-4	0.0547	536	0.1790
GRU	2	100	0.1	1e-4	0.0551	523	0.1904
GRU	2	100	0.01	1e-4	0.0529	1212	0.1825
GRU	2	120	0.1	1e-4	0.0531	1283	0.1820
GRU	3	80	0.01	1e-4	0.0533	760	0.1802
GRU	3	100	0.1	1e-4	0.0533	432	0.1818
LSTM	2	80	0.01	1e-4	0.0524	263	0.1641
LSTM	2	80	0.01	1e-3	0.0508	224	0.1613
LSTM	3	100	0.01	1e-4	0.0532	436	0.1732
LSTM	2	80	0.005	1e-2	0.0509	273	0.1663
LSTM	2	100	0.005	1e-4	0.053	763	0.1724
LSTM	2	60	0.005	1e-4	0.0478	1287	0.2789

Figure 30: Grid Search Parameters for tuning

We chose the parameters located in the fifth row from the bottom, i.e. `cell type`: LSTM, `num layers`: 2, `beta end`: 1e-3, `lr`: 1e-3, which was able to achieve a low CRPS as well as a low number of invalid constructions.